How Far from Clinical Deployment? Evaluating the Complete Unsupervised Domain Adaptation Pipeline in Medical Imaging

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Abstract

Deploying unsupervised domain adaptation (UDA) in clinical practice requires choosing which algorithm to use and which of its trained models to ship. However, the deployment (target) domain is unlabeled, so models cannot be evaluated directly on it, leaving it unclear which to select. We address this by evaluating the complete UDA pipeline, considering both adaptation and label-free selection together. Our study covers eleven clinically relevant cross-domain scenarios from nine medical imaging datasets, with ten UDA algorithms and 13 label-free selection methods (validators), evaluating over 80,000 trained models in total. By this, we find that a capable adapted model usually exists, but identifying it without target labels is difficult: the validator-selected models leave a large and structural target performance gap to the best available one, with no evaluated validator consistently reliable. Towards closing it, we explore two strategies, ensembling and a small target-labeling budget; both narrow this gap but do not close it entirely. Overall, deployable UDA depends on the complete pipeline; addressing the less explored selection step could bring much of current UDA closer to clinical use. Our code is open sourced at Complete UDA Pipeline.

1. Introduction

Deep learning models excel at medical image-based diagnosis, but their performance often degrades substantially under domain shifts arising from differences in imaging modalities, acquisition protocols, and patient populations [23]. Annotating training data from a new domain is costly, requiring scarce medical expertise. Unsupervised domain adaptation (UDA) addresses both challenges by transferring knowledge from a labeled source domain to an unlabeled target domain, with recent algorithms adapting robustly across clinically relevant shifts, such as cross-modality [50], cross-site [27], and cross-age [11]. However, deploying UDA in clinical routine requires a practitioner to decide which algorithm to use and which of its checkpoints (a model snapshot at a training iteration) to ship, as shown in Figure 1. Thus, model selection is an unavoidable part of the complete UDA pipeline. Furthermore, without target labels, candidate models cannot be evaluated directly on the target domain, leaving it unclear which one to select for deployment: the standard approach of comparing them on a held-out labeled validation set is unavailable [34]. Yet, existing medical UDA surveys [5, 14, 23, 25, 39, 42, 51] and evaluations [3, 8, 36, 41, 44] primarily focus on adaptation in isolation, leaving the selection step largely unexamined. As a result, it remains unclear how the complete UDA pipeline behaves under clinical deployment conditions, and thus how far current UDA truly is from clinical deployment.

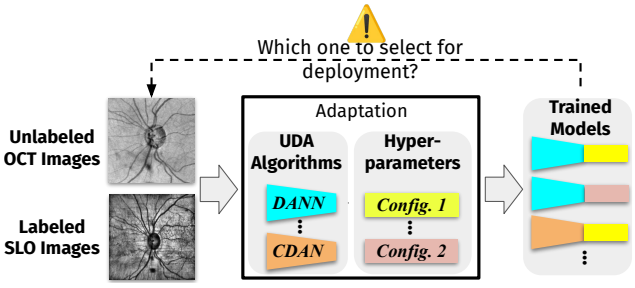


Figure 1. The complete UDA pipeline at deployment. Adaptation trains models on a labeled source domain (here SLO) and an unlabeled target domain (here OCT), producing many candidate models that vary in UDA algorithm and hyperparameter configuration. At deployment, one model needs to be selected, yet without target labels they cannot be evaluated directly to decide which.

To address this, we evaluate the complete UDA pipeline in medical imaging, considering both adaptation and label-free selection that clinical deployment requires. For adaptation, we include ten UDA algorithms spanning multiple paradigms, with some tailored to medical imaging [15, 54]. For selection, we adopt 13 established label-free selection approaches (a.k.a. validators), such as IWCV [43] and DEV [53]. Each validator assigns a scalar validation score to each checkpoint without using target labels, and

the checkpoint with the best score is selected. Our evaluation covers brain MRI, chest X-ray (CXR), and retinal imaging, with eleven clinically relevant cross-domain scenarios in total. Brain MRI and CXR use four datasets each, with within-modality shifts arising from differences in hospitals, scanners, or patient populations, while retinal imaging involves across-modality shifts between SLO and OCT. Altogether, this amounts to roughly 16,500 checkpoint configurations (each set by the algorithm, its hyperparameters, and training iteration), or over 80,000 trained checkpoints once repeated across folds or random seeds.

Our study points to a consistent picture across different cross-domain scenarios. Adaptation itself is often effective: a capable adapted model usually exists among the candidates. What proves difficult is identifying it without target labels: the models selected by the evaluated validators leave a large target performance gap to the best available model. We further trace the origin of this gap and find it largely structural: no single evaluated validator works consistently well across all scenarios; one that succeeds in some fails in others, so no default choice can be recommended in advance. In the absence of a consistently reliable validator across all scenarios, towards closing the gap, we turn to two strategies. Instead of relying on one validator, the first ensembles predictions across validator-selected checkpoints from every algorithm; instead of remaining fully label-free, the second spends a small target-labeling budget. Both narrow the gap, though neither closes it entirely and each carries its own limitations.

Our contributions are summarized as follows:

- We present, to our knowledge, the first study to evaluate the complete adaptation and label-free selection UDA pipeline in medical imaging under clinical deployment conditions, spanning ten algorithms, 13 validators, and nine datasets across eleven cross-domain scenarios.
- By considering adaptation and selection as a whole, we find that a capable adapted model usually exists, yet the remaining distance to clinical deployment lies largely in label-free selection: the selection gap is large and structural, with no evaluated validator consistently reliable.
- Towards closing the gap, we point to two strategies, ensembling and a small target-labeling budget; both narrow this gap and open directions for more reliable selection.

2. Related Work

2.1. Surveys and Benchmarks on Medical UDA

UDA in medical imaging has been reviewed from several angles. Choudhary et al. [5] give an early overview of deep domain adaptation, Guan and Liu [14] provide a widely used taxonomy across modalities, and Kumari and Singh [23] survey more recent deep UDA methods and datasets. More recent surveys cover emerging diffusion- and foundation-model approaches [51], broader reviews of distribution shift also include UDA [42], and others focus on specific settings such as functional brain data [39] and segmentation [25]. Alongside these surveys, several benchmarks evaluate adaptation under realistic shifts: M3DA [41] for 3D MRI/CT segmentation, CrossMoDA [8] for cross-modality segmentation, M3-UDA [36] for multi-hospital fetal ultrasound detection, and Chamarthi et al. [3] and Sultana et al. [44] for skin lesion classification. These efforts have established rigorous comparisons of adaptation methods on clinically relevant data, but the selection step remains largely underexplored. Our study complements them by evaluating the complete pipeline that clinical deployment requires, adaptation and label-free selection together.

2.2. Label-Free Model Selection in UDA

Selecting a model without target labels has been studied mainly in general-vision UDA, where a range of validators have been proposed [12, 17, 35, 38, 43, 48, 52, 53] and evaluated. Musgrave et al. [34] provide one of the first UDA model selection benchmarks covering many algorithms, though over a relatively small set of three label-free validators. Subsequent evaluations [10, 18, 24] strengthen evaluation practice further with more validators, with SKADA-Bench [24] additionally covering broader modalities, though with an emphasis on shallow adaptation methods. These works, however, primarily focus on selecting checkpoints within individual algorithms. In practice, deployment also requires choosing which algorithm to use, so a more realistic setting also calls for selecting across algorithms. Beyond this, medical images pose distinct challenges, exhibiting more pronounced and varied shifts than natural images [23], under which the behavior of these validators remains unclear. To address these, our study examines selection also across algorithms, over a broad set of 13 validators and ten deep UDA algorithms, including medical-specific ones, and, in particular, under the shifts that clinical routine faces, such as cross-modality, cross-institution, and cross-cohort, assessing how the complete UDA pipeline behaves under deployment conditions.

3. Datasets and Methods

We study the complete UDA pipeline in medical imaging under the conditions a clinical deployment would face, focusing on classification tasks such as disease diagnosis. The pipeline comprises two stages. In the adaptation stage, UDA algorithms share a common structure, a backbone, a classification head, and an adaptation module, and optimize a joint objective

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \, \mathcal{L}_{\text{adapt}}, \tag{1}$$

where $\mathcal{L}_{\text{cls}}$ supervises the model with source labels, $\mathcal{L}_{\text{adapt}}$ aligns feature distributions across domains, and λ controls

Table 1. Overview of the datasets, UDA algorithms, and validators used in our study. **(a)** Datasets across brain MRI, CXR, and retinal imaging (SLO / OCT), with the number of positive (Pos.) and negative (Neg.) samples per dataset. Positive and negative denote, respectively, Alzheimer's disease and cognitively normal for brain MRI, pneumonia and non-pneumonia for CXR, and glaucoma and non-glaucoma for retinal imaging. **(b)** UDA algorithms spanning multiple paradigms. **(c)** Validators grouped into source-guided and target-based criteria.

(a) Datasets

Organ	Modality	Dataset	Pos.	Neg.
Brain	MRI	ADNI-1 [19]	200	221
		ADNI-2 [19]	159	232
		ADNI-3 [19]	85	431
		AIBL [9]	78	477
Chest	X-Ray	RSNA [49]	6,012	20,672
		Child CXR [22]	4,273	1,583
		LDD [2]	5,776	3,919
		CRD [1]	9,237	10,319
Eye	SLO	FairDomain [45]	4,453	5,547
	OCT		4,453	5,547

(b) UDA Algorithms

Feat. Dist.: MMD [30]	Pseudo Lab.: ATDOC [26]
Info. Max: MCC [20]	Cls. Disc.: MCD [37]
SVD Loss: BNM [6]	Adv. Align: DANN [13], CDAN [31], DALN [4]
Medical-Spec.: AD2A (Brain MRI) [15], CoUDA (CXR) [54]	

(c) Validators

Source-Guided:	Source-Risk [12], IWCV [43], DEV [53], DEV-N [35]
Target-Based:	Entropy [33], InfoMax [34], Corr-C [48], BNM (V) [35], MCC (V) [20], SND [38], ClassAMI [35], MixVal [17], TransScore [52]

the adaptation strength. Running an algorithm with a given λ produces a training run, during which we save a sequence of checkpoints. Each checkpoint θ is thus uniquely defined by its algorithm, its adaptation strength λ, and its training iteration, and every checkpoint is a candidate for deployment.

In the selection stage, we replicate the constraint of clinical deployment: target labels are unavailable, so the model to deploy must be chosen without them. The role of a validator is to provide a label-free validation score for the target performance. Formally, a validator is a scoring function

$$V : \theta \mapsto s \in \mathbb{R}, \tag{2}$$

that maps a checkpoint θ to a scalar score $s = V(\theta)$ without using target labels. The selected model for deployment is the checkpoint with the best validation score,

$$\theta^\star = \arg\max_{\theta \in \Theta} V(\theta), \tag{3}$$

where Θ is the candidate checkpoint pool and scores are oriented so that higher is better.

Datasets. To construct medical UDA scenarios, four widely adopted brain MRI datasets are used: ADNI-1, ADNI-2, ADNI-3 [19], and AIBL [9]. Subjects appearing in multiple ADNI datasets are kept in only one to prevent data leakage. In addition, four publicly available CXR datasets are used: RSNA [49], Child CXR [22], LDD [2], and CRD [1]. For retinal imaging, we use FairDomain dataset [45], which provides paired SLO and OCT acquisitions of the same cohort. For brain MRI and CXR, each dataset is treated as a separate domain, and transfer is performed across datasets within each modality; for retinal data, transfer is performed between SLO and OCT modalities. Brain MRI datasets contain Alzheimer's disease and cognitively normal subjects, CXR datasets contain pneumonia and non-pneumonia subjects, and retinal datasets contain glaucoma and non-glaucoma subjects. All modalities undergo standard preprocessing following prior work [15, 45, 54]. Dataset statistics are summarized in Table 1 (a).

Methods. Since we study the complete pipeline as a whole, we include a representative set of UDA algorithms and, crucially, a diverse set of label-free validators. As shown in Table 1 (b) and (c), the ten algorithms span multiple paradigms, including feature-distance minimization [30], adversarial alignment [4, 13, 31], information maximization [20], SVD loss [6], pseudo-labeling [26], classifier discrepancy [37], and medical-specific techniques [15, 54]. The 13 validators cover both source-guided criteria [12, 35, 43, 53] and target-based ones [17, 20, 33, 34, 38, 48, 52].

Experimental Setup. Following Guan et al. [15], five UDA scenarios (source→target) are constructed for brain MRI: ADNI-1→ADNI-2, ADNI-1→ADNI-3, ADNI-2→ADNI-1, ADNI-2→ADNI-3, and ADNI-1+2→AIBL. Following Feng et al. [11] and Liu et al. [28], two CXR transfers are used, each in both directions: RSNA↔Child CXR and LDD↔CRD. For retinal imaging, the two cross-modality directions SLO→OCT and OCT→SLO are evaluated. Both source and target data are split into training and validation sets, and target performance is measured on the target validation set using balanced accuracy. For datasets without predefined splits, stratified five-fold cross-validation is performed; for datasets with explicit train/test splits, results are averaged over three random seeds. In both cases the mean $\pm$ standard deviation is reported; median values with 95% confidence intervals are additionally provided in the supplementary material.

Implementation Details. For all experiments, the classification head is a three-layer MLP with a dropout rate of 0.5. For brain MRI, a 3D ResNet-50 [16] trained from scratch is used as the backbone, with a batch size of eight per domain. For CXR and retinal imaging, a 2D ResNet-50 pretrained on ImageNet [7] is used, with a batch size of 48 per domain. All algorithms are trained with AdamW (weight decay 1e-4) and a one-cycle learning rate schedule with warm-up and

Table 2. Complete UDA results on ADNI-1→ADNI-2 (target accuracy, %; eight best validators by **Across-Algo** accuracy shown). Each cell reports the target accuracy of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	88.2±2.0	89.3±4.0	90.0±4.0	89.9±3.0	91.0±3.5	91.0±1.7	89.9±4.4	85.7±9.8	87.4±4.3	92.0±0.93	89.5±2.0	93.2±2.0	90.9±2.4
DEV-N	80.9±5.8	81.7±2.8	84.9±2.0	81.8±8.4	80.8±9.1	83.5±3.3	82.9±3.4	76.0±12	81.0±6.2	87.6±1.9	-	85.1±1.3	-
Source-Risk	81.2±5.8	82.1±3.1	81.8±4.0	81.1±8.0	81.1±9.0	83.4±3.3	84.1±4.1	73.2±12	81.5±6.3	86.8±2.8	-	84.4±2.5	-
InfoMax	84.2±2.1	84.8±5.7	87.3±4.4	86.0±1.5	88.2±3.8	81.8±9.6	83.5±6.9	78.9±10	83.3±5.6	86.3±2.2	-	83.2±7.2	-
BNM (V)	84.7±2.1	84.8±5.7	87.5±4.0	84.9±3.8	87.2±2.2	81.8±9.6	83.5±6.9	78.9±10	83.3±5.6	86.3±2.2	-	81.8±9.6	-
ClassAMI	80.9±6.4	60.0±7.3	82.1±3.1	82.7±3.8	84.6±3.5	83.3±2.4	79.8±2.0	79.7±8.2	59.2±9.8	81.6±2.1	-	81.0±2.3	-
SND	73.9±12	83.4±4.1	80.0±2.6	79.2±5.8	85.5±4.9	78.2±5.9	78.5±8.6	75.8±12	82.2±5.6	80.7±8.7	-	80.7±8.7	-
MCC (V)	84.2±2.7	84.4±6.4	84.9±3.3	84.5±4.0	86.4±3.7	80.4±8.3	85.9±3.6	80.7±13	83.1±3.7	87.1±3.7	-	80.4±2.3	-
Corr-C	84.6±4.1	80.4±4.3	83.4±3.6	83.1±3.6	84.2±6.7	77.8±5.3	80.4±3.4	78.3±8.8	84.1±4.4	85.6±5.1	-	79.6±3.0	-

a peak learning rate of 1e-3, for 10k iterations on brain MRI and retinal imaging and 30k iterations on CXR. The adaptation strength is varied over $\lambda \in \{0.1, 0.5, 1.0\}$, giving three runs per algorithm. After warm-up, checkpoints are saved at uniform intervals, yielding 50 checkpoints per run and 150 checkpoints per algorithm. $\mathcal{L}_{\text{cls}}$ is binary cross-entropy with class-balanced weighting derived from the source labels. All runs are conducted on an A6000 GPU.

4. Results and Discussion

We use ADNI-1→ADNI-2 as an illustrative example for the complete UDA pipeline in Table 2, which reports the target accuracy of the checkpoint each validator selects. This selection is compared against the *Oracle*, which selects directly with target labels. We define the *selection gap* as the difference in target accuracy between the *Oracle*-selected model and the model selected by a label-free validator. The per-algorithm columns report selection within a single algorithm, and **Avg.** is their mean, while **Across-Algo** pools all algorithms' checkpoints and selects across them. Due to space constraints, we display the eight best validators by their **Across-Algo** accuracy. Full results for all scenarios can be found in the supplementary material. We report *SourceOnly* (no adaptation) and *TargetOnly* (trained on labeled target data) as references.

4.1. Adaptation Works in Principle

As shown in Table 2, under *Oracle* selection, ADNI-1→ADNI-2 scenario reaches 93.2% accuracy when pooling all checkpoints (**Across-Algo**), exceeding both *SourceOnly* (88.2%) and *TargetOnly* (90.9%). The gain is broad: the *Oracle*-selected checkpoint beats *SourceOnly* for seven of the nine algorithms, with a per-algorithm average of 89.5% (**Avg.**), and surpasses *TargetOnly* for three.

Figure 2 extends Table 2 to all eleven UDA scenarios, reporting the same four quantities under *Oracle* selection. At the per-algorithm level, the average (**Avg.**) exceeds *SourceOnly* in eight of the eleven scenarios, and in two it even surpasses *TargetOnly*. Pooling checkpoints across algorithms strengthens this further: **Across-Algo** lies above *SourceOnly* in every scenario, and exceeds *TargetOnly* in four. Averaged over all scenarios (**Mean**), the ordering is consistent: *SourceOnly* < **Avg.** < **Across-Algo**, with **Across-Algo** reaching 84.5% and approaching *TargetOnly* (87.5%). Adaptation is therefore often effective across these clinical UDA scenarios; in particular, a capable adapted model exists in most scenarios (evidenced by **Across-Algo**). Nevertheless, this model is only useful if it can be identified without target labels, as deployment requires.

4.2. Large Selection Gap from Validators

The capable model established above can only be deployed if a validator selects it. Return to ADNI-1→ADNI-2 scenario as shown in Table 2, under **Across-Algo**, the *Oracle* attains 93.2%. Over the same across-algorithm pool, the model selected by the best validator (DEV-N) reaches 85.1%, and that selected by the best algorithm–validator pair (DALN with InfoMax) reaches 88.2%. Even these best cases fall 5.0 to 8.1 points short of the *Oracle*.

Figure 3 extends Table 2 to all eleven UDA scenarios, reporting the target accuracy of the model selected by the best validator over the across-algorithm pool (**Best Validator**), by the best algorithm–validator pair (**Best Pair**), and by the across-algorithm *Oracle*. In every scenario, both **Best Validator** and **Best Pair** fall below the *Oracle*: the gap reaches up to 10.6 points (RSNA→Child CXR) and averages 6.4 points across scenarios (**Mean**). Crucially, **Best Validator** and **Best Pair** differ from scenario to scenario and are themselves identified using target labels; at deployment, where no labels are available to choose them in advance, the realized gap can only be larger. A capable model exists in the pool, but selecting it without labels leaves a substantial portion of the achievable performance unrealized.

The Selection Gap Persists Across Architectures. To test whether our findings depend on the backbone, we repeat RSNA→Child CXR with three additional architec-

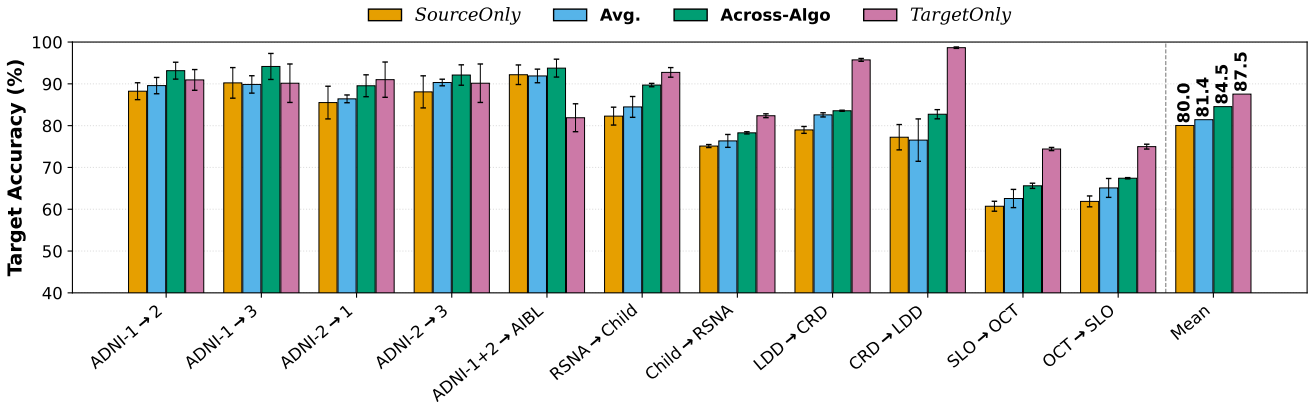


Figure 2. Target accuracy under *Oracle* selection across UDA scenarios. Bars show *SourceOnly* (no adaptation), the per-algorithm average (**Avg.**), **Across-Algo** (selection over all algorithms' pooled checkpoints), and *TargetOnly* (labeled-target). **Across-Algo** exceeds *SourceOnly* in all scenarios and approaches *TargetOnly*, indicating a capable adapted model usually exists. Error bars: standard deviation.

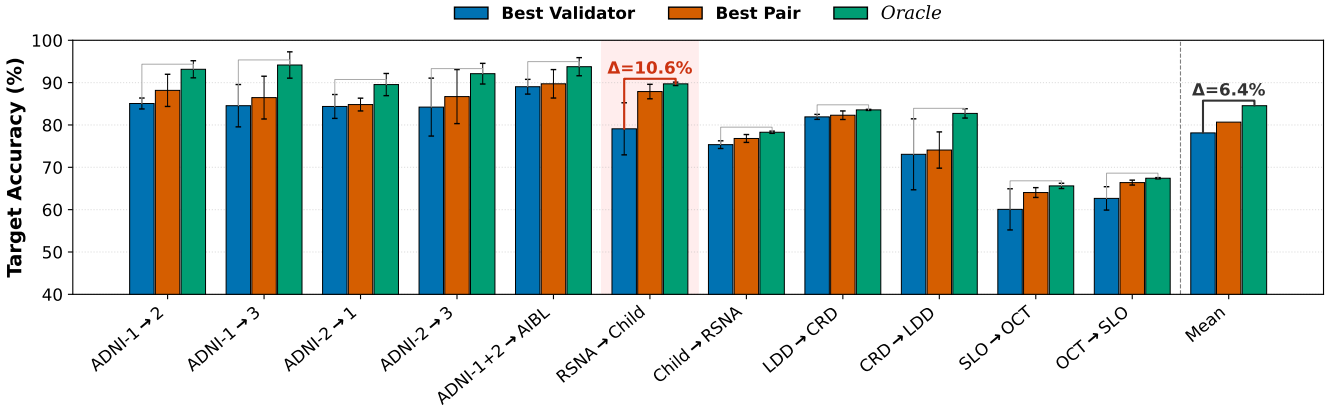


Figure 3. Target accuracy across UDA scenarios under three selections (%). Bars show the accuracy of the model selected by the best validator over the across-algorithm pool (**Best Validator**), by the best algorithm–validator combination (**Best Pair**), and by the across-algorithm *Oracle*. The first two fall below the *Oracle* in every scenario, with the gap up to 10.6 points and averaging 6.4 (**Mean**). Note that **Best Validator** and **Best Pair** differ from scenario to scenario and are identified using target labels, so they cannot be chosen in advance; the gap realized at deployment can only be larger. Error bars: standard deviation.

Table 3. Target accuracy (%) on RSNA→Child CXR for four backbones. **Best Val.** is the accuracy of the model selected by the best validator under the across-algorithm pool, and *Oracle* that of the actual best model. The gap persists across all four backbones.

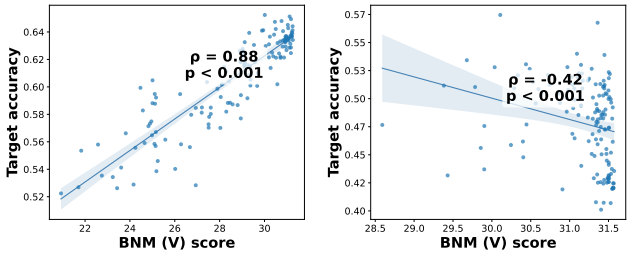
Backbone	Best Val.	*Oracle*	Δ*Oracle*
ResNet-50	79.1	89.7	10.6
ResMLP	70.9	79.2	8.3
ConvNeXt	82.6	90.2	7.6
DeiT	80.4	86.5	6.1

tures spanning distinct families: ConvNeXt [29] (a modern CNN), ResMLP [47] (an MLP-based model), and DeiT [46] (a vision transformer). As shown in Table 3, the model selected by the best validator leaves a noticeable gap to the *Oracle* for every architecture. This suggests the selection gap is not specific to a particular backbone.

4.3. Structural Selection Gap from Validators

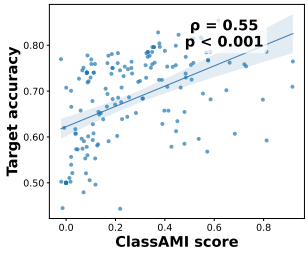
Having established the magnitude of the selection gap, we now investigate its origin. The gap may arise either from a suboptimal choice of validator, which a more suitable one would remedy, or from a more structural limitation that no evaluated validator overcomes. Since selection is fundamentally a ranking problem (a validator succeeds by ranking checkpoints so that the best one scores highest), we trace the selection gap to validator reliability, measured by the Spearman rank correlation ρ between validation scores and true target accuracy. For validators where a lower score indicates a better checkpoint, the sign is flipped so that a positive correlation always denotes the intended direction.

Within-Algorithm Selection. Figure 4 (a) and (b) fix the scenario OCT→SLO and validator BNM (V) and vary the algorithm: BNM (V) ranks DALN's checkpoints well and

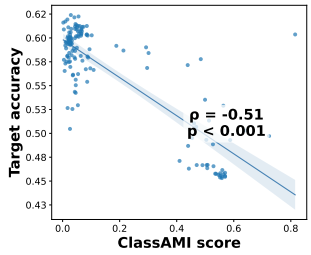


(a) OCT→SLO, DALN

(b) OCT→SLO, ATDOC



(c) MMD, ADNI-2→ADNI-1



(d) MMD, SLO→OCT

Figure 4. Within-algorithm Spearman correlation (ρ) between validation score and target accuracy. Each point is a checkpoint. The same validator can flip from reliable to reversed with a change of algorithm (a, b) or scenario (c, d) alone.

in the designed direction ($\rho = 0.88, p < 0.001$), yet ranks ATDOC's checkpoints in reverse ($\rho = -0.42, p < 0.001$). Figure 4 (c) and (d) fix the algorithm (MMD) and validator (ClassAMI) and vary the scenario: the correlation is positive as intended for ADNI-2→ADNI-1 ($\rho = 0.55, p < 0.001$) but reversed for SLO→OCT ($\rho = -0.51, p < 0.001$). Further individual analyses can be found in the supplementary material. Figure 5 aggregates the within-algorithm Spearman correlation between validation scores and target accuracy two ways: (a) averaged over algorithms for a per-scenario view, and (b) averaged over scenarios for a per-algorithm view; in both, blue denotes correlation in the intended direction (darker is stronger) and warm red the opposite. On average, validators are moderately informative, but their reliability is uneven: no validator attains consistent correlation across all scenarios in (a) or all algorithms in (b), and the strongest validator differs from column to column. Source-Risk, for example, is the most reliable validator for ADNI-2→ADNI-3 ($\rho = 0.68$) yet drops to near zero for Child CXR→RSNA ($\rho = 0.06$); likewise, InfoMax is strongest for AD2A ($\rho = 0.80$) but falls to $\rho = 0.24$ for ATDOC.

Across-Algorithm Selection. Figure 6 (a) shows Source-Risk on ADNI-1→ADNI-2 for a single algorithm (BNM): within this algorithm, it ranks checkpoints reasonably well ($\rho = 0.57, p < 0.001$). Figure 6 (b) instead pools checkpoints across all algorithms for the same scenario and validator. To form this pool while keeping the comparison meaningful, we take, for each algorithm, the ten checkpoints that the validator scores highest: this retains the competitive checkpoints worth selecting among, while ten per algorithm provides a sufficient sample to estimate the rank correlation. On this pooled set, the correlation collapses to near zero ($\rho = -0.04, p = 0.678$). Figure 7 reports the same across-algorithm correlation across all validators and scenarios: most validators are unreliable or contradict their intended direction, with correlations weak or negative.

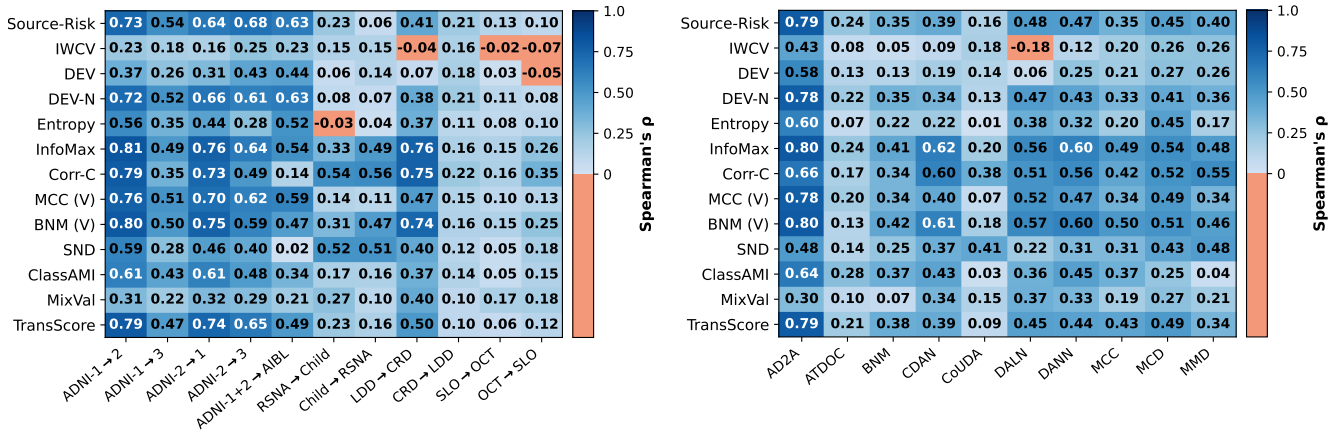
For within-algorithm selection, a suitable validator may exist among evaluated ones, but which one differs by algorithm and scenario, so it cannot be chosen in advance. For across-algorithm selection, even this fails: most evaluated validators correlate weakly or in reverse. The gap therefore appears largely structural rather than a matter of choosing a better validator among those evaluated ones.

4.4. Strategies to Narrow the Selection Gap

The structural selection gap leaves a large distance between what adaptation achieves and what label-free selection realizes at deployment. In the absence of a validator consistently reliable across scenarios, we explore two practical strategies and show that each recovers part of this gap.

Ensembling. Since no single evaluated validator is consistently reliable, we avoid committing to one and instead ensemble the checkpoints selected by different validators, similar to Hu et al. [18] but additionally ensembling across algorithms. In the across-algorithm setting, where checkpoints from all algorithms are pooled, we select one checkpoint per algorithm with every validator and ensemble these selections by averaging their predictions. As shown in Table 4, ensembling matches or exceeds **Best Val.** in most scenarios. Again **Best Val.** is the model selected by the best validator for each scenario, which differs across scenarios and cannot be known in advance. The remaining gap to the *Oracle* is small in several cases (e.g., 1.0 for OCT→SLO and 2.3 for LDD→CRD), showing that it recovers much of the achievable performance. It nonetheless remains far from the *Oracle* in other cases such as RSNA→Child CXR.

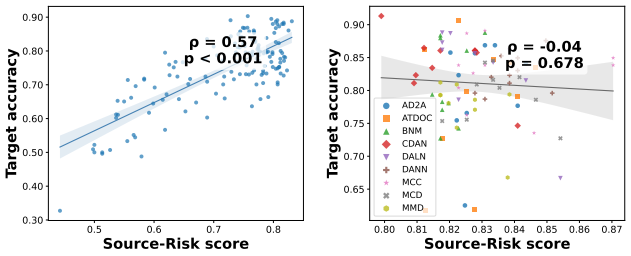
Small Target Labeling Budget. Since fully label-free selection cannot be trusted under the evaluated validators, we explore whether spending a small budget on target labeling can identify a better checkpoint. We label a small subset of the target validation set and select the checkpoint with the highest accuracy on it. The budget is varied over five levels (B_1 to B_5): 5, 10, ..., 25 labeled samples for brain MRI, whose target validation set is around 100 samples, and 1% to 5% for the others. Each budget is repeated over ten random trials, and we report the mean. As shown in Table 4, a small budget improves selection steadily, surpassing both **Best Val.** (78.1) and ensembling (78.0) from B_4 (20 labeled samples for brain MRI, 4% for the other scenarios) on average, though a gap to the *Oracle* still remains.



(a) Per-scenario view (averaged over algorithms)

(b) Per-algorithm view (averaged over scenarios)

Figure 5. Within-algorithm Spearman correlation (ρ) between validation scores and target accuracy. Blue denotes correlation in the intended direction (darker is stronger), warm red the opposite. No validator is consistently reliable across all scenarios or algorithms, and the most reliable one differs from column to column.



(a) Within BNM algorithm

(b) Across Algorithms

Figure 6. Spearman correlation (ρ) between Source-Risk scores and target accuracy on ADNI-1→ADNI-2. Each point is a checkpoint. (a) Within BNM algorithm, the correlation is moderate. (b) Across all algorithms, it collapses to near zero.

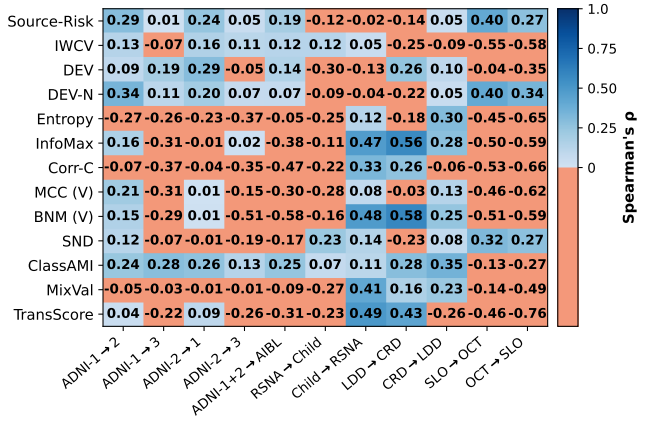


Figure 7. Across-algorithm Spearman correlation (ρ) between validation scores and target accuracy. Blue denotes correlation in the intended direction (darker is stronger), warm red the opposite. Most cells are weak or negative, showing that most validators are unreliable in this across-algorithm setting.

4.5. Discussion

Our results show that the main barrier between current UDA and clinical deployment lies less in producing a capable adapted model, which usually exists, than in selecting it without target labels. Under *Oracle* selection, such a model exists in the **Across-Algo** pool in most scenarios, often recovering much of the gap between *SourceOnly* and *TargetOnly* and at times exceeding both, confirming that adaptation is usually effective. The evaluated validators, however, do not reliably identify this model, leaving a large and structural gap to the best available one. For within-algorithm selection, a reliable validator may exist, but which one is best shifts from algorithm to algorithm and from scenario to scenario, so no fixed choice can be made in advance. For across-algorithm selection, most validators correlate only weakly with target performance, and many rank checkpoints opposite to their intended direction. Note that we do not optimize *TargetOnly* extensively; in several scenarios it falls below the per-algorithm *Oracle* average (**Avg.**) or even *SourceOnly*, which may stem from training on the target domain alone, with less labeled data than the combined source and target signal exploited by adaptation, and from target class imbalance (e.g., AIBL).

The unreliability of evaluated validators also explains an observation that might otherwise question the value of adaptation: under many validators, the selected adapted model performs worse than the *SourceOnly* baseline (e.g., MCC with InfoMax in Table 2). We interpret this as a selection problem, not adaptation alone: a capable adapted model often exists (the *Oracle* confirms it), but an unreliable validator may fail to find it and sometimes picks one worse than

Table 4. Two strategies for narrowing the selection gap (target accuracy, %). **Ens.** ensembles the checkpoints selected by all validators across all algorithms. **Small Labeling Budget** selects the best checkpoint on a small labeled target subset, at five levels (B_1 to B_5; 5–25 samples for brain MRI, 1–5% otherwise). On average, ensembling matches **Best Val.** (the best validator under the across-algorithm pool), and the labeling budget surpasses both from around B_4, though a gap to the *Oracle* remains for both.

	Ens.	**Small Labeling Budget**					**Best Val.**	*Oracle*
Scenario		B_1	B_2	B_3	B_4	B_5		
ADNI-1→ADNI-2	89.3	78.6	83.0	84.0	86.5	87.5	85.1	93.2
ADNI-1→ADNI-3	87.3	78.7	80.8	81.1	81.2	83.3	84.5	94.2
ADNI-2→ADNI-1	83.2	80.4	79.1	81.5	82.6	84.0	84.4	89.5
ADNI-2→ADNI-3	87.2	79.5	79.9	83.5	81.3	83.6	84.2	92.1
ADNI-1+2→AIBL	89.4	81.6	82.9	82.9	85.1	86.0	89.0	93.8
RSNA→Child CXR	79.8	76.6	78.4	81.5	83.1	85.0	79.1	89.7
Child CXR→RSNA	74.4	74.3	75.8	76.3	76.9	76.9	75.3	78.3
LDD→CRD	81.3	80.2	81.6	82.0	82.2	82.1	81.9	83.6
CRD→LDD	56.9	75.9	78.1	80.2	79.7	80.4	73.1	82.7
SLO→OCT	63.1	58.9	61.2	60.8	61.3	62.6	60.1	65.6
OCT→SLO	66.4	60.0	60.9	63.5	64.5	64.4	62.7	67.4
Mean	78.0	75.0	76.5	77.9	78.6	79.6	78.1	84.5

not adapting. On the other hand, in Table 2, under *Oracle* selection ATDOC and MCD fall below *SourceOnly*, so the limitation here lies also in adaptation. This reflects either our relatively small hyperparameter set (λ and the training iteration) or the unsuitability of these algorithms for this transfer. It nonetheless reinforces the need for reliable validators: at the across-algorithm selection scale, a good validator should recognize and avoid such negative transfer.

We also notice that validator reliability, measured by Spearman correlation ρ, and the target accuracy of the selected model do not always align. As shown in Figure 7 and the **Across-Algo** column of Table 2, for ADNI-1→ADNI-2, ClassAMI has a higher correlation than InfoMax ($\rho = 0.24$ vs. 0.16) yet selects a less accurate model (81.0% vs. 83.2%). This is because the two measure different things: ρ captures how well a validator orders all checkpoints, whereas selection depends on its single top-scored checkpoint, so a validator with lower ρ can still place a good checkpoint at the top. This suggests two complementary needs: a reliable validator to produce a good ranking, and potentially a further step to identify the truly best one.

How Far from Clinical Deployment? The answer is two-sided. Under *Oracle* selection, a capable model exists in the checkpoint pool in most scenarios, so adaptation itself is often effective. What is missing is the ability to identify that model without target labels. Deployment therefore depends on the complete pipeline, not adaptation alone: selection remains a largely unsolved step, and closing it would bring much of current UDA closer to clinical use. In some scenarios the achievable accuracy itself remains low, and for some algorithms even the *Oracle*-selected model falls below *SourceOnly*, so better adaptation algorithms are still needed. Yet even there, and for any future algorithm, a capable model still has to be identified. Improving algorithms raises what is achievable, while reliable selection determines what is actually reached at deployment.

The two strategies in Section 4.4 recover part of this selection gap but do not close it, and both carry practical costs. Ensembling requires training potentially multiple algorithms across their hyperparameters, since which suits a given transfer is not known in advance, and multiple forward passes per image at inference; it is computationally costly and can even harm performance under naive aggregation when poor checkpoints are included (e.g., CRD→LDD). The labeling budget, though small in relative terms (5 to 25 volumes for brain MRI, 1% to 5% otherwise), carries a real annotation cost: even a single brain MRI volume is time-consuming, and 1% of a large dataset can amount to hundreds of images (e.g., over 260 for RSNA), more so for richer tasks such as segmentation or detection. Active selection [21, 32, 40] may match these gains with fewer labels. Overall, reliable label-free selection remains the central open problem for clinically deployable UDA.

Limitations and Future Work. Our study covers binary classification; extending it to multi-class, multi-label, segmentation, or detection tasks, where the prediction structure differs and selection criteria may need adapting, is a valuable future direction. As classification lies at the core of these more complex tasks, label-free selection may not become easier there, and the gap could even widen. We evaluate by balanced accuracy, while other metrics such as sensitivity and specificity also matter clinically, and whether validators rank checkpoints well under them remains open.

5. Conclusion

In this work, we evaluated the complete UDA pipeline in medical imaging, adaptation and label-free selection together, under clinical deployment conditions. Spanning eleven clinically relevant cross-domain scenarios from nine datasets across brain MRI, CXR, and retinal imaging, ten UDA algorithms, and 13 validators, our evaluation covers over 80,000 trained checkpoints. By considering adaptation and selection as a whole, we find that a capable adapted model often exists, but selecting it without target labels is difficult: the validator-selected models leave a large and structural gap to the best available one, as no evaluated validator is consistently reliable, for either within-algorithm or across-algorithm selection. Two strategies, ensembling and a small target-labeling budget, narrow this gap but do not close it entirely, and both have limitations. Overall, deployable UDA depends on both adaptation and selection working together; addressing the less explored selection step could bring much of current UDA closer to clinical use.

Acknowledgments

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6. Supplementary Material

In this supplementary material, we provide the full results of our complete unsupervised domain adaptation (UDA) pipeline for each evaluated clinically relevant cross-domain scenario in Section 6.1. In addition, we provide the full within-algorithm validator reliability analyses, reported per algorithm and per scenario, in Section 6.2.

6.1. Full Results on the Complete UDA Pipeline

This section provides the full per-scenario complete UDA results. Each table reports, for one cross-domain scenario, the target accuracy of the checkpoint selected by every validator (rows) for every algorithm (columns), along with *Oracle*, *SourceOnly*, and *TargetOnly* references. For each scenario, the first table reports mean±std and the immediately following table reports median with 95% CI (e.g., Table 5 is mean±std and Table 6 is median with 95% CI, and analogously for the remaining ten scenarios). Tables 5–25 cover the eleven clinically relevant transfer scenarios. Brain MRI: ADNI-1→ADNI-2 (Table 5), ADNI-1→ADNI-3 (Table 7), ADNI-2→ADNI-1 (Table 9), ADNI-2→ADNI-3 (Table 11), ADNI-1+2→AIBL (Table 13). Chest X-ray: RSNA→Child CXR (Table 15), Child CXR→RSNA (Table 17), LDD→CRD (Table 19), CRD→LDD (Table 21). Retinal: OCT→SLO (Table 23), SLO→OCT (Table 25).

We also assess whether adaptation works in principle and the size of the selection gap using the median with 95% confidence intervals, shown in Figures 8 and 9. The conclusions match those drawn from the mean ± standard deviation in the main paper.

6.2. Full Validator Reliability Analyses for Within-Algorithm Selection

This section provides the full per-scenario validator reliability analyses. Each table reports, for one cross-domain scenario, the within-algorithm Spearman correlation (ρ) between each validation score and target accuracy, computed over the checkpoints of each algorithm. Positive values always indicate ranking in the intended direction. The ρ values here are means across folds or random seeds, whereas the example scatter plots of the main paper show a single fold or random seed for readability, so individual cells may not match exactly. Tables 27–37 cover the eleven clinically relevant transfer scenarios. Brain MRI: ADNI-1→ADNI-2 (Table 27), ADNI-1→ADNI-3 (Table 28), ADNI-2→ADNI-1 (Table 29), ADNI-2→ADNI-3 (Table 30), ADNI-1+2→AIBL (Table 31). Chest X-ray: RSNA→Child CXR (Table 32), Child CXR→RSNA (Table 33), LDD→CRD (Table 34), CRD→LDD (Table 35). Retinal: OCT→SLO (Table 36), SLO→OCT (Table 37).

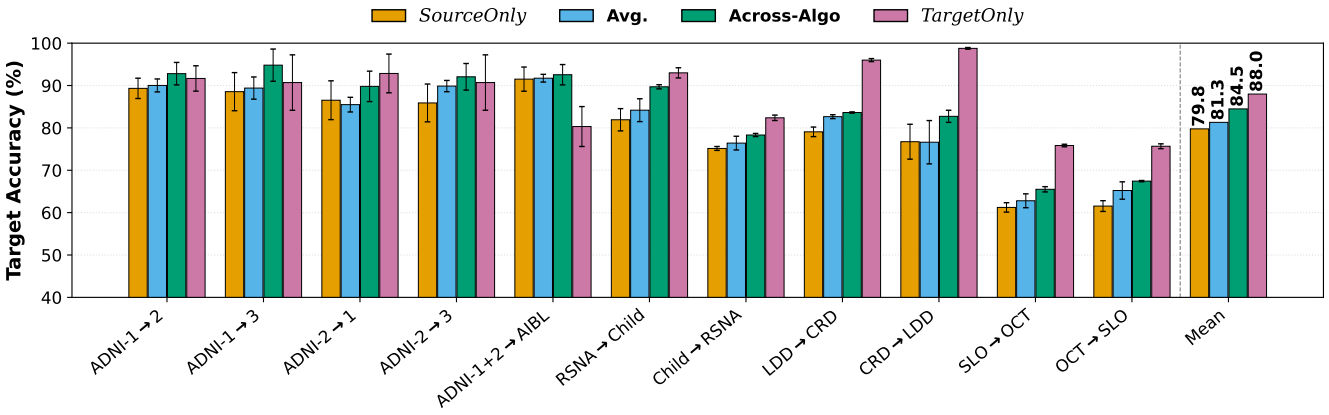


Figure 8. Target accuracy under *Oracle* selection across UDA scenarios, reported as the median with 95% confidence intervals. Bars show *SourceOnly* (no adaptation), the per-algorithm average (**Avg.**), **Across-Algo** (selection over all algorithms' pooled checkpoints), and *TargetOnly* (labeled-target). **Across-Algo** exceeds *SourceOnly* in all scenarios and approaches *TargetOnly*, indicating a capable adapted model usually exists. Error bars: 95% confidence interval.

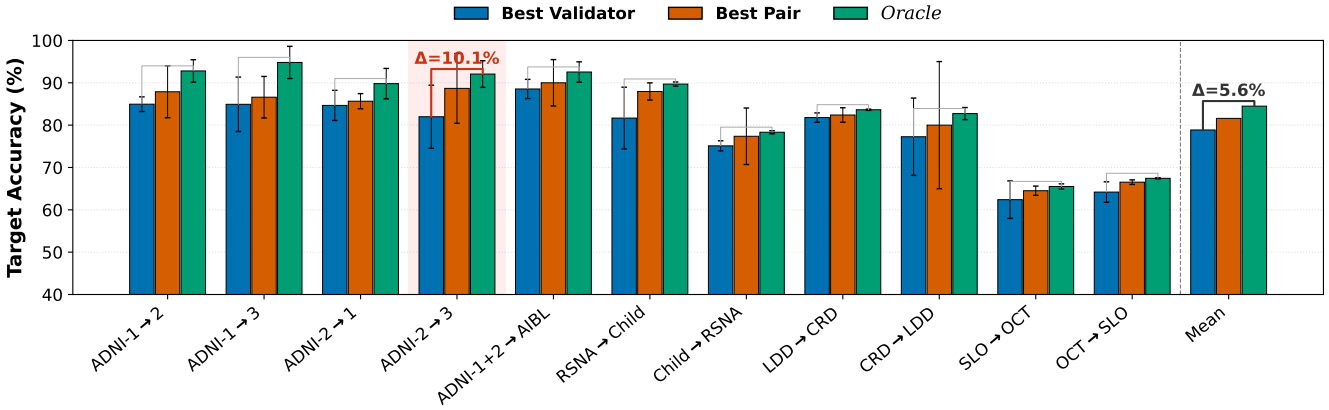


Figure 9. Target accuracy across UDA scenarios under three selections (%), reported as the median with 95% confidence intervals. Bars show the accuracy of the model selected by the best validator over the across-algorithm pool (**Best Validator**), by the best algorithm–validator combination (**Best Pair**), and by the across-algorithm *Oracle*. The first two fall below the *Oracle* in every scenario, with the gap up to 10.1 points and averaging 5.6 (**Mean**). **Best Validator** and **Best Pair** differ from scenario to scenario and are identified using target labels, so they cannot be chosen in advance; the gap realized at deployment can only be larger. Error bars: 95% confidence interval.

Table 5. Complete UDA results on ADNI-1→ADNI-2 (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	88.2±2.0	89.3±4.0	90.0±4.0	89.9±3.0	91.0±3.5	91.0±1.7	89.9±4.4	85.7±9.8	87.4±4.3	92.0±0.93	89.5±2.0	93.2±2.0	90.9±2.4
Source-Risk	81.2±5.8	82.1±3.1	81.8±4.0	81.1±8.0	81.1±9.0	83.4±3.3	84.1±4.1	73.2±12	81.5±6.3	86.8±2.8	-	84.4±2.5	-
IWCV	76.8±9.7	80.3±5.3	74.6±12	81.1±6.4	80.1±8.4	76.0±10	76.4±3.8	72.3±11	80.1±11	84.3±2.8	-	76.7±4.4	-
DEV	78.0±11	82.2±3.2	84.5±6.2	82.0±6.7	84.9±6.6	76.9±11	75.9±7.3	70.3±15	68.6±19	81.4±8.2	-	79.4±13	-
DEV-N	80.9±5.8	81.7±2.8	84.9±2.0	81.8±8.4	80.8±9.1	83.5±3.3	82.9±3.4	76.0±12	81.0±6.2	87.6±1.9	-	85.1±1.3	-
Entropy	60.9±15	58.8±20	56.7±15	57.0±16	50.0±15	60.2±15	57.5±16	57.9±13	70.1±19	56.9±15	-	50.0±10	-
InfoMax	84.2±2.1	84.8±5.7	87.3±4.4	86.0±1.5	88.2±3.8	81.8±9.6	83.5±6.9	78.9±10	83.3±5.6	86.3±2.2	-	83.2±7.2	-
Corr-C	84.6±4.1	80.4±4.3	83.4±3.6	83.1±3.6	84.2±6.7	77.8±5.3	80.4±3.4	78.3±8.8	84.1±4.4	85.6±5.1	-	79.6±3.0	-
MCC (V)	84.2±2.7	84.4±6.4	84.9±3.3	84.5±4.0	86.4±3.7	80.4±8.3	85.9±3.6	80.7±13	83.1±3.7	87.1±3.7	-	80.4±2.3	-
BNM (V)	84.7±2.1	84.8±5.7	87.5±4.0	84.9±3.8	87.2±2.2	81.8±9.6	83.5±6.9	78.9±10	83.3±5.6	86.3±2.2	-	81.8±9.6	-
ClassAMI	80.9±6.4	60.0±7.3	82.1±3.1	82.7±3.8	84.6±3.5	83.3±2.4	79.8±2.0	79.7±8.2	59.2±9.8	81.6±2.1	-	81.0±2.3	-
SND	73.9±12	83.4±4.1	80.0±2.6	79.2±5.8	85.5±4.9	78.2±5.9	78.5±8.6	75.8±12	82.2±5.6	80.7±8.7	-	80.7±8.7	-
MixVal	75.9±12	57.4±6.4	77.9±8.7	69.8±13	66.8±16	82.1±8.7	71.3±16	70.2±15	71.8±16	60.0±16	-	61.1±18	-
TransScore	85.2±2.8	82.7±4.8	86.9±2.2	85.9±2.0	79.3±14	81.4±8.2	76.4±9.7	76.9±12	83.3±5.4	87.8±3.1	-	78.5±11	-

Table 6. Complete UDA results on ADNI-1→ADNI-2 (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 5. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	AD2A	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	89.3±2.4	88.6±5.4	89.1±5.2	89.8±3.7	91.7±4.8	91.3±1.9	90.3±4.8	92.3±1.2	89.3±12	87.7±5.4	90.0±1.5	92.8±2.6	91.7±3.0
Source-Risk	82.4±7.9	82.2±3.8	82.0±4.8	82.7±10.0	82.0±12	84.4±4.1	81.4±4.1	86.7±3.1	74.6±14	81.1±7.5	-	84.4±3.0	-
IWCV	75.8±13	80.7±7.0	70.7±14	80.7±8.4	83.7±10	79.2±12	76.5±4.8	86.9±7.5	74.6±14	85.8±13	-	79.2±5.2	-
DEV	83.3±13	82.2±3.8	83.7±8.1	83.7±8.4	87.5±7.0	80.8±13	79.4±8.9	81.6±9.2	66.0±15	69.7±21	-	81.6±17	-
Entropy	53.2±17	50.0±0.85	50.0±17	50.0±18	50.0±0.0	50.0±17	50.0±18	50.0±17	50.0±19	77.3±19	-	50.0±0.0	-
InfoMax	85.0±2.5	82.3±7.2	86.5±6.0	86.8±1.9	86.7±4.7	83.3±12	84.3±8.5	85.7±2.7	84.5±12	81.1±7.4	-	83.3±9.0	-
Corr-C	85.0±4.5	82.5±4.9	82.3±4.3	82.7±4.7	87.3±7.7	79.3±6.5	82.0±4.2	87.9±6.1	81.7±11	81.4±5.0	-	79.3±3.8	-
MCC (V)	84.8±3.7	84.2±8.6	84.5±4.3	85.6±4.9	86.7±4.6	83.3±11	86.7±4.5	87.5±4.7	85.2±17	81.4±4.7	-	83.3±11	-
BNM (V)	85.7±2.5	82.3±7.2	86.5±5.2	86.8±4.6	86.7±2.4	83.3±12	84.3±8.5	85.7±2.7	84.5±12	81.1±7.4	-	83.3±12	-
SND	79.0±14	83.3±4.8	79.7±3.6	75.9±6.8	84.5±6.3	78.8±8.0	79.6±11	82.8±5.4	81.7±14	83.4±7.6	-	79.7±10	-
ClassAMI	82.5±8.0	62.1±9.6	81.1±4.1	83.0±4.9	83.9±4.5	83.3±3.3	79.4±2.6	81.1±2.7	84.4±9.7	58.5±12	-	81.1±2.8	-
DEV-N	80.7±7.9	82.2±3.4	85.2±2.8	82.7±10.0	82.0±12	84.4±4.1	81.4±4.1	86.9±2.4	74.6±14	78.8±7.1	-	85.0±1.7	-
MixVal	80.3±15	55.7±8.5	78.0±11	71.3±16	65.2±18	83.2±10	77.7±18	68.8±20	67.6±19	79.8±20	-	53.0±19	-
TransScore	85.7±3.6	82.4±6.7	86.9±2.7	86.4±2.5	85.8±17	83.3±11	79.9±12	87.7±3.5	81.9±15	81.1±7.0	-	83.2±15	-

Table 7. Complete UDA results on ADNI-1→ADNI-3 (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	90.2±3.7	87.3±4.5	90.7±3.1	93.5±3.5	90.8±4.1	90.8±3.3	89.0±5.6	86.9±5.8	88.6±3.5	91.1±4.3	89.9±2.0	94.2±3.1	90.2±4.6
Source-Risk	82.8±8.3	78.0±9.9	83.8±5.3	82.7±8.8	82.4±7.2	77.6±6.2	78.0±7.1	77.9±10	82.9±5.1	83.0±6.7	-	83.5±4.8	-
IWCV	70.0±10	76.7±11	79.6±4.9	84.1±5.9	74.7±15	74.0±12	56.5±9.1	69.3±11	79.2±6.7	75.3±12	-	64.9±15	-
DEV	73.9±12	69.4±11	82.2±4.9	86.5±5.0	81.1±7.2	79.9±6.5	62.9±13	72.8±11	79.4±7.1	81.6±5.8	-	84.0±7.5	-
DEV-N	83.3±7.1	79.2±9.7	83.8±5.3	82.7±8.8	82.4±5.0	77.3±5.7	76.1±8.5	76.6±12	75.2±9.8	83.2±4.4	-	84.5±5.0	-
Entropy	67.8±13	66.8±15	57.2±15	57.3±14	63.2±18	66.1±17	64.0±11	64.1±18	62.5±17	50.3±0.72	-	50.0±0.0	-
InfoMax	79.0±5.9	72.0±14	79.7±5.0	78.2±4.1	79.4±6.8	70.6±7.9	72.7±6.5	72.4±12	80.2±5.8	80.2±5.7	-	70.6±7.9	-
Corr-C	79.7±6.7	70.8±5.1	76.8±1.5	75.9±3.7	74.5±4.3	71.3±8.3	69.9±5.0	67.1±13	76.1±4.8	72.1±11	-	68.7±8.5	-
MCC (V)	79.1±7.9	74.9±14	84.6±2.3	84.5±4.0	79.9±6.3	75.9±9.8	72.4±4.2	73.8±18	85.1±3.9	83.5±7.6	-	75.9±9.8	-
BNM (V)	78.2±7.3	75.0±15	79.7±5.0	78.2±4.1	79.4±6.8	70.6±7.9	72.7±6.5	74.3±13	80.9±4.9	80.2±5.7	-	70.6±7.9	-
ClassAMI	80.7±6.5	54.6±18	83.6±2.0	82.9±2.2	80.6±6.7	81.6±5.3	81.3±7.1	74.2±9.8	58.0±11	80.1±7.3	-	80.6±3.6	-
SND	73.8±6.0	71.6±5.0	77.5±4.5	75.4±6.0	76.8±4.3	73.2±5.3	71.2±9.0	69.4±8.8	77.4±3.2	74.5±2.0	-	76.2±3.9	-
MixVal	78.2±11	64.9±18	73.3±9.1	78.1±12	66.7±14	82.3±4.0	72.6±14	72.7±11	53.4±6.0	59.8±12	-	64.7±13	-
TransScore	76.3±6.4	69.9±13	81.7±2.7	81.1±5.9	76.9±5.0	69.6±7.1	47.3±19	73.9±13	75.0±6.0	72.9±22	-	62.6±16	-

Table 8. Complete UDA results on ADNI-1→ADNI-3 (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 7. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	AD2A	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	88.5±4.5	86.2±5.1	90.0±3.5	93.8±4.7	91.6±5.5	90.7±4.4	87.8±6.9	90.8±5.0	86.0±7.4	87.6±4.6	89.4±2.6	94.8±3.8	90.7±6.5
Source-Risk	82.8±9.7	79.2±10	85.4±6.7	80.1±11	82.3±9.8	78.5±8.3	79.1±8.7	84.7±8.2	80.5±13	84.9±6.3	-	82.2±6.4	-
IWCV	64.3±11	78.9±15	81.5±5.3	86.1±7.6	77.8±19	79.4±15	56.0±11	75.5±17	69.8±14	76.1±7.6	-	60.5±16	-
DEV	73.1±14	73.6±14	81.8±6.7	86.1±6.5	79.5±9.7	83.0±7.8	60.5±17	81.0±7.8	74.7±14	76.1±8.1	-	84.1±9.7	-
Entropy	68.4±17	64.3±19	50.0±17	50.0±16	51.2±20	64.5±19	69.6±13	50.0±0.81	66.7±23	51.0±17	-	50.0±0.0	-
InfoMax	77.0±7.3	73.3±18	80.8±6.6	79.7±5.0	77.4±8.7	72.4±11	75.0±8.2	79.3±6.8	76.5±14	78.4±6.9	-	72.4±11	-
Corr-C	79.0±8.0	72.6±6.3	76.7±1.8	76.7±4.0	76.0±5.2	72.4±11	70.7±6.3	74.6±14	72.6±14	77.9±6.1	-	71.6±11	-
MCC (V)	77.0±10	80.7±18	84.2±3.1	86.0±4.5	76.8±7.4	72.2±12	72.8±5.3	79.3±7.9	78.1±23	86.6±4.9	-	72.2±12	-
BNM (V)	75.5±8.9	82.4±18	80.8±6.6	79.7±5.0	77.4±8.7	72.4±11	75.0±8.2	79.3±6.8	78.1±17	78.4±5.2	-	72.4±11	-
SND	74.2±7.9	72.6±6.2	78.7±5.5	74.3±7.5	77.1±5.4	74.9±6.7	69.3±11	74.0±2.3	72.8±11	77.9±4.1	-	77.1±5.2	-
ClassAMI	84.2±7.6	53.2±23	84.4±2.2	82.2±2.5	78.5±8.6	80.5±6.3	78.2±7.5	79.2±7.8	77.5±13	55.6±14	-	80.2±4.2	-
DEV-N	82.8±7.8	77.6±11	85.4±6.7	80.1±11	80.8±6.4	78.5±7.6	78.0±11	84.7±4.5	80.5±16	77.9±11	-	84.9±6.4	-
MixVal	80.9±14	57.7±18	70.8±11	80.7±17	69.4±16	81.6±4.9	75.4±17	54.8±15	74.8±15	56.1±7.2	-	65.3±16	-
TransScore	74.3±8.4	71.7±18	81.1±3.0	82.5±8.0	77.4±6.7	71.9±9.2	40.1±23	80.1±26	78.1±16	77.9±6.4	-	70.9±20	-

Table 9. Complete UDA results on ADNI-2→ADNI-1 (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	85.5±3.9	86.6±3.1	86.6±3.7	87.4±2.8	87.2±3.2	85.5±3.8	86.9±3.5	85.0±4.1	85.2±5.3	87.2±3.1	86.4±0.88	89.5±2.6	91.0±4.2
Source-Risk	77.4±5.0	80.4±4.7	78.8±6.9	75.2±7.5	79.8±5.4	80.2±6.3	78.7±5.0	76.7±4.3	77.5±8.4	80.8±4.7	-	78.5±4.7	-
IWCV	75.1±3.6	79.5±3.5	79.0±5.2	79.2±7.3	76.5±6.0	77.7±6.1	80.8±2.4	75.3±7.7	70.0±9.8	79.5±6.1	-	81.9±4.4	-
DEV	76.3±2.8	79.5±3.5	80.7±5.6	77.8±8.4	81.2±3.4	77.7±6.1	78.1±3.2	73.8±5.7	69.6±10.0	80.5±4.1	-	78.1±3.5	-
DEV-N	78.8±3.1	79.7±5.0	79.4±6.8	78.2±6.5	78.9±5.9	82.8±3.9	80.5±3.4	77.0±4.5	76.8±9.0	84.8±1.5	-	84.4±2.8	-
Entropy	53.9±6.1	56.6±14	61.7±16	64.2±19	50.5±1.2	67.5±18	68.7±17	66.0±15	70.0±14	56.9±16	-	56.1±14	-
InfoMax	81.0±4.3	79.1±6.9	83.2±3.5	80.7±4.4	81.5±2.2	82.0±4.4	79.5±6.0	79.7±6.4	76.3±8.2	84.1±5.5	-	82.0±4.4	-
Corr-C	79.6±5.5	80.3±4.5	81.2±5.4	80.0±5.1	81.5±3.5	80.8±5.9	77.2±5.0	80.4±5.2	77.0±8.4	83.8±5.4	-	80.8±5.9	-
MCC (V)	80.5±4.5	81.3±4.7	82.2±3.9	82.4±3.9	82.5±3.5	73.6±11	78.5±4.5	80.1±6.3	77.4±7.2	84.1±3.7	-	73.6±11	-
BNM (V)	80.9±4.1	79.3±6.8	83.2±3.5	80.7±4.4	82.0±3.1	82.0±4.4	80.2±6.3	80.5±6.6	76.3±8.2	83.8±5.5	-	82.0±4.4	-
ClassAMI	76.3±3.1	77.7±2.0	81.5±4.0	80.5±4.1	80.8±3.8	80.4±3.9	78.2±5.3	78.1±5.4	64.4±15	79.0±5.9	-	80.0±4.2	-
SND	81.9±7.2	77.5±8.0	75.7±1.9	77.4±6.1	78.8±4.6	67.5±7.8	75.8±8.0	75.0±3.1	74.6±3.6	72.3±1.4	-	75.9±8.7	-
MixVal	83.2±6.5	73.9±12	71.0±13	82.3±4.0	72.7±17	73.4±14	78.9±7.8	75.0±13	60.8±16	68.8±14	-	63.3±11	-
TransScore	78.3±5.6	79.4±3.5	82.4±3.2	80.5±4.7	81.8±3.3	77.2±3.5	79.1±6.5	77.8±4.0	77.2±7.0	82.2±3.4	-	78.4±5.0	-

Table 10. Complete UDA results on ADNI-2→ADNI-1 (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 9. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	AD2A	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	86.5±4.6	85.4±3.7	85.9±5.0	87.5±3.7	87.0±4.6	84.6±4.8	87.0±3.9	85.7±3.4	84.1±5.2	82.0±5.7	85.5±1.7	89.8±3.6	92.8±4.6
Source-Risk	76.9±6.8	81.2±6.3	75.1±8.2	76.2±9.9	78.7±6.9	80.2±7.6	77.5±6.0	81.1±5.7	78.5±5.1	72.6±9.2	-	78.0±5.7	-
IWCV	75.5±5.0	78.6±4.5	76.0±5.7	78.0±9.7	76.5±7.5	75.5±8.0	80.2±3.0	79.9±8.3	74.3±8.9	66.9±13	-	80.2±5.3	-
DEV	76.1±3.6	78.6±4.5	80.0±7.0	77.5±12	81.8±3.9	75.5±8.0	78.9±4.1	79.4±5.2	74.9±6.4	66.5±13	-	78.5±4.5	-
Entropy	50.0±7.0	50.0±16	50.0±15	52.3±19	50.0±1.3	67.2±20	79.4±16	50.0±17	73.8±14	70.6±19	-	50.0±15	-
InfoMax	81.0±4.9	78.5±9.0	83.3±4.8	80.2±5.3	82.0±2.8	79.9±5.2	77.5±7.9	84.7±7.0	77.5±8.3	75.3±11	-	79.9±5.2	-
Corr-C	77.4±6.7	78.3±5.2	84.5±6.2	78.6±6.2	81.0±4.5	79.8±8.1	75.7±6.0	84.5±6.7	78.6±6.3	76.2±12	-	79.8±8.1	-
MCC (V)	82.0±5.7	81.8±6.2	80.4±4.8	80.4±4.8	81.6±4.4	67.2±12	79.4±5.6	83.2±5.0	78.8±8.3	75.3±9.3	-	67.2±12	-
BNM (V)	81.0±4.7	78.5±9.0	83.3±4.8	80.2±5.3	82.0±4.0	79.9±5.2	77.5±7.9	83.2±7.0	77.5±8.3	75.3±11	-	79.9±5.2	-
SND	83.1±9.6	77.6±11	76.2±2.2	79.8±7.5	78.8±5.7	65.5±10	72.0±8.2	72.5±1.8	75.2±4.0	75.1±4.7	-	79.4±10	-
ClassAMI	76.9±3.6	78.6±2.5	80.7±5.4	80.3±5.3	81.6±5.1	79.8±5.4	76.4±6.3	79.6±7.2	78.1±7.2	64.3±20	-	79.3±5.7	-
DEV-N	77.4±3.7	79.8±6.3	75.8±8.1	77.6±8.8	77.1±7.2	83.3±5.0	80.0±3.7	85.7±1.8	78.8±5.3	70.8±9.5	-	84.7±3.6	-
MixVal	85.3±8.1	71.3±15	75.0±13	81.1±5.0	82.2±17	79.2±15	81.1±10	70.0±14	78.8±17	53.7±20	-	58.1±13	-
TransScore	77.1±6.5	78.5±4.4	84.4±3.4	79.8±5.8	80.9±4.5	77.4±4.4	77.5±8.5	82.2±4.4	79.7±5.0	78.5±9.6	-	78.6±6.8	-

Table 11. Complete UDA results on ADNI-2→ADNI-3 (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	88.1±3.8	89.3±3.8	91.1±3.3	91.2±3.0	90.7±2.9	90.1±4.1	90.1±2.9	89.2±3.1	90.0±3.0	91.2±2.8	90.3±0.74	92.1±2.4	90.2±4.6
Source-Risk	82.3±7.3	81.2±5.7	80.8±6.8	84.9±5.9	82.5±2.8	82.6±6.4	79.2±7.4	81.3±5.4	83.3±6.0	85.2±6.4	-	82.7±7.9	-
IWCV	81.0±2.3	79.6±2.2	82.0±5.8	84.4±6.2	84.0±7.0	81.2±2.9	78.8±8.0	77.4±7.6	82.0±5.8	85.2±5.5	-	84.2±6.8	-
DEV	81.6±7.7	78.2±2.7	86.4±3.9	83.2±5.2	84.9±6.3	78.8±6.6	82.0±6.5	77.3±4.0	81.7±6.8	83.2±5.2	-	82.8±6.8	-
DEV-N	82.0±7.2	80.7±3.1	81.1±7.3	84.1±6.5	83.7±3.4	83.2±5.8	80.7±5.7	80.2±6.8	82.8±5.7	85.5±5.9	-	78.3±9.7	-
Entropy	57.0±16	56.6±15	65.8±22	58.4±19	49.9±0.26	50.0±0.0	64.2±20	69.0±18	67.7±16	51.9±4.3	-	50.0±0.0	-
InfoMax	81.5±4.8	83.2±5.4	83.5±4.9	82.7±4.7	78.9±6.3	81.7±3.8	80.1±3.0	82.6±7.0	78.5±8.7	84.8±6.5	-	80.6±3.2	-
Corr-C	74.8±10	77.5±2.8	74.0±4.7	76.7±1.5	72.3±6.6	76.0±4.9	76.9±2.7	72.4±2.6	79.2±7.0	79.7±5.4	-	76.9±2.7	-
MCC (V)	83.8±6.7	82.7±5.3	86.7±6.4	85.6±5.8	84.6±6.8	84.7±6.3	82.4±6.4	78.6±7.2	79.9±7.6	74.4±19	-	82.8±3.9	-
BNM (V)	81.2±5.2	83.2±5.4	85.3±1.3	82.7±4.7	78.9±6.3	81.7±3.8	80.1±3.0	54.4±7.5	60.9±17	83.9±6.0	-	54.4±7.5	-
ClassAMI	81.6±6.9	82.2±4.8	82.0±5.4	82.4±4.4	83.4±7.4	81.5±6.3	83.0±5.1	79.1±6.1	67.7±9.9	84.8±4.3	-	81.6±5.6	-
SND	76.0±5.1	77.7±2.8	74.2±4.4	71.7±3.9	74.9±8.2	74.8±2.0	75.4±2.2	76.5±4.5	78.5±5.5	75.3±9.9	-	71.2±5.7	-
MixVal	83.8±6.1	63.9±13	66.5±16	75.2±13	70.6±16	75.6±13	65.9±18	75.8±15	65.9±17	71.0±18	-	59.0±12	-
TransScore	82.0±4.4	84.2±6.2	83.5±4.9	80.8±4.8	83.2±8.2	80.9±5.0	78.8±6.4	74.1±7.0	78.1±6.0	78.4±18	-	79.8±4.4	-

Table 12. Complete UDA results on ADNI-2→ADNI-3 (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 11. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	AD2A	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	85.9±4.5	89.3±5.0	92.0±3.9	90.8±4.1	90.3±3.7	90.3±5.6	88.9±3.7	90.7±3.4	87.5±3.6	89.1±3.5	89.9±1.3	92.1±3.1	90.7±6.5
Source-Risk	84.7±9.5	80.8±7.0	77.8±7.8	84.2±8.3	80.8±3.0	80.6±8.0	79.5±10	88.0±7.4	83.9±6.5	85.0±7.5	-	80.8±11	-
IWCV	81.1±2.9	80.8±2.7	79.6±6.1	84.3±7.7	80.1±8.1	81.9±3.8	77.2±10	88.0±6.0	80.7±9.0	80.6±6.9	-	81.9±8.9	-
DEV	80.9±11	77.5±3.3	88.1±4.7	83.2±6.6	84.8±7.7	78.7±7.8	80.0±7.3	80.4±6.0	77.5±5.5	84.2±6.9	-	79.6±8.2	-
Entropy	50.0±17	50.0±17	50.0±22	50.0±21	50.0±0.29	50.0±0.0	50.0±20	50.0±4.8	63.4±20	71.3±19	-	50.0±0.0	-
InfoMax	80.2±6.1	84.8±6.7	85.6±6.0	81.9±5.9	77.8±7.2	82.2±4.2	79.5±3.7	84.2±7.5	83.4±9.3	74.7±8.8	-	81.4±3.7	-
Corr-C	77.6±13	79.0±3.4	75.6±5.7	76.5±1.8	73.5±9.0	75.5±6.1	77.3±3.0	81.4±6.8	72.4±3.5	81.0±8.5	-	77.3±3.0	-
MCC (V)	81.8±7.9	82.6±6.8	88.7±8.2	83.0±7.0	84.4±9.2	84.2±8.3	81.4±7.9	80.2±25	79.8±9.7	76.8±8.4	-	81.4±4.3	-
BNM (V)	80.2±6.9	84.8±6.7	85.6±1.5	81.9±5.9	77.8±7.2	82.2±4.2	79.5±3.7	84.2±7.5	53.6±10	65.7±22	-	53.6±10	-
SND	76.1±7.0	77.5±3.8	76.7±5.2	71.1±5.1	76.9±11	75.0±2.3	76.6±2.6	80.0±13	77.3±5.9	80.0±7.3	-	72.8±7.5	-
ClassAMI	81.5±9.4	81.0±6.1	84.4±5.9	81.7±5.9	83.5±10	80.5±6.9	82.1±6.5	86.0±4.7	76.5±7.5	69.5±13	-	82.0±7.5	-
DEV-N	84.7±9.5	80.8±4.3	77.8±8.7	84.1±8.3	84.8±3.9	81.9±7.1	79.5±7.3	88.0±6.8	82.8±8.4	83.8±6.9	-	77.2±13	-
MixVal	82.1±7.7	56.9±15	59.5±19	77.4±17	73.1±19	79.5±16	58.8±18	69.8±20	78.9±19	56.7±20	-	56.7±15	-
TransScore	80.2±5.4	86.8±7.7	85.6±6.0	80.1±5.9	83.9±9.7	82.2±6.2	81.4±8.1	83.6±23	73.0±9.1	78.2±7.6	-	81.4±5.6	-

Table 13. Complete UDA results on ADNI-1+2→AIBL (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	92.2±2.3	91.6±2.0	92.8±1.1	93.4±1.7	93.1±2.0	91.8±1.6	91.4±1.9	88.0±6.0	91.9±1.7	93.1±2.6	91.9±1.5	93.8±2.1	81.9±3.3
Source-Risk	88.4±3.0	85.0±3.5	86.7±4.5	87.8±3.9	85.6±8.2	87.6±2.7	84.0±3.2	79.9±4.9	86.2±4.6	86.5±2.0	-	84.0±5.1	-
IWCV	86.8±3.3	83.2±4.6	83.8±4.7	84.3±6.0	84.2±8.0	86.4±4.5	83.3±3.2	82.3±4.2	83.2±3.9	87.8±1.9	-	84.4±3.3	-
DEV	87.3±2.7	84.2±5.8	83.4±5.2	82.2±12	89.2±1.7	85.4±4.8	87.4±2.2	82.5±4.5	82.1±4.3	79.5±5.0	-	89.0±1.7	-
DEV-N	87.6±4.2	85.0±3.5	85.0±5.0	88.9±3.5	85.6±8.2	88.2±2.3	84.0±3.2	79.9±4.9	84.6±4.8	88.8±3.7	-	82.7±5.0	-
Entropy	88.3±3.3	84.5±5.4	88.6±4.8	86.9±8.0	66.0±20	66.7±15	79.0±16	80.7±7.8	79.1±16	80.5±18	-	64.8±16	-
InfoMax	84.7±6.7	82.4±4.9	78.9±3.2	84.2±5.0	79.5±7.2	79.4±5.5	81.6±5.0	82.2±6.4	85.8±4.2	82.9±2.9	-	81.6±5.0	-
Corr-C	80.4±6.8	77.3±4.7	75.4±3.0	77.5±3.6	77.6±5.8	74.3±4.7	75.0±3.4	77.3±3.2	77.4±3.6	75.9±3.9	-	75.1±3.2	-
MCC (V)	88.3±3.8	86.0±3.0	84.7±5.5	89.7±3.4	86.3±4.6	80.6±11	83.8±2.9	81.3±6.2	84.8±3.1	88.8±3.8	-	84.1±4.8	-
BNM (V)	88.2±4.1	84.7±3.0	80.2±1.1	83.9±5.2	82.1±8.4	79.4±5.5	81.6±5.0	65.7±8.2	74.6±13	86.7±6.3	-	65.7±8.2	-
ClassAMI	84.9±7.1	68.7±19	83.9±5.5	84.0±5.1	86.0±5.6	85.9±2.4	84.5±2.3	82.8±5.4	74.2±13	84.8±2.6	-	83.3±3.4	-
SND	74.0±1.8	75.2±3.8	76.0±5.6	77.9±3.6	76.6±6.8	75.2±4.7	74.1±5.7	74.8±2.0	78.3±4.4	73.4±4.4	-	76.9±5.2	-
MixVal	88.8±4.0	59.4±12	82.0±7.5	84.1±8.7	77.8±12	76.5±16	81.2±6.1	79.4±5.5	76.9±14	67.9±13	-	70.9±16	-
TransScore	88.5±4.0	82.6±5.8	79.5±3.9	82.1±6.7	80.1±8.3	80.7±6.0	85.0±1.3	80.5±5.9	82.1±4.0	85.1±3.4	-	80.1±4.7	-

Table 14. Complete UDA results on ADNI-1+2→AIBL (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 13. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	AD2A	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	91.5±2.9	91.8±2.1	92.5±1.4	92.5±1.9	92.0±1.9	91.5±1.9	91.3±2.5	92.0±3.2	89.6±7.9	92.3±2.1	91.7±0.91	92.5±2.4	80.3±4.7
Source-Risk	86.9±3.6	86.5±3.5	87.4±5.5	88.6±5.3	86.6±9.7	87.4±3.0	85.3±3.9	86.5±2.4	78.7±6.0	87.1±5.5	-	85.1±6.6	-
IWCV	86.9±4.1	84.0±5.9	85.1±6.3	85.2±7.6	85.1±11	84.9±6.0	84.2±3.8	88.5±2.4	83.3±5.5	82.8±5.5	-	84.9±4.2	-
DEV	87.4±3.0	86.0±6.8	84.8±6.8	87.0±15	89.1±2.3	84.8±6.4	87.0±2.8	81.8±6.3	83.7±6.0	82.5±5.2	-	88.5±2.3	-
Entropy	89.9±3.4	86.4±6.7	90.0±5.5	88.7±10	53.9±22	61.5±18	85.3±20	85.1±22	83.7±10	86.8±19	-	61.5±18	-
InfoMax	85.8±9.3	84.4±6.2	79.7±4.2	83.8±6.6	80.5±9.0	81.6±6.7	83.5±5.7	82.3±3.7	84.3±7.2	86.8±5.2	-	83.5±5.7	-
Corr-C	76.6±7.4	78.9±6.2	76.6±3.6	77.1±4.0	79.1±7.4	74.0±6.0	75.5±4.7	75.3±5.3	77.8±3.2	78.7±4.5	-	76.0±4.1	-
MCC (V)	87.9±4.3	86.8±3.5	84.3±6.9	88.4±4.2	87.7±5.8	85.6±13	84.8±3.8	88.2±4.9	77.8±6.4	85.4±3.8	-	85.6±6.2	-
BNM (V)	86.3±4.6	83.5±3.4	79.9±1.4	83.8±6.6	84.9±11	81.6±6.7	83.5±5.7	86.9±7.0	62.1±9.4	79.7±16	-	62.1±9.4	-
SND	74.4±2.1	76.0±5.1	76.3±7.1	77.8±4.5	73.0±8.0	74.0±6.3	73.0±7.1	73.2±5.8	75.0±2.1	78.8±5.8	-	75.5±6.4	-
ClassAMI	84.1±9.1	78.9±20	85.6±7.2	85.5±6.6	83.9±6.6	85.5±3.2	84.4±2.8	84.4±3.5	84.4±6.3	77.7±18	-	83.7±3.9	-
DEV-N	86.9±5.6	86.5±3.5	82.8±6.1	88.7±4.3	86.6±9.7	87.7±3.0	85.3±3.9	87.7±4.8	78.7±6.0	83.4±5.7	-	81.8±6.6	-
MixVal	90.6±4.9	57.7±14	79.7±8.2	86.6±11	82.5±14	82.8±19	79.9±7.6	61.5±15	77.2±6.6	81.1±18	-	75.5±19	-
TransScore	87.4±4.6	82.3±7.5	79.7±5.2	82.5±8.2	80.5±11	83.5±7.1	84.9±1.7	86.7±4.3	77.8±6.9	80.6±4.8	-	78.9±5.0	-

Table 15. Complete UDA results on RSNA→Child CXR (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	82.3±2.1	83.1±1.8	82.8±2.0	84.7±1.6	83.5±1.4	87.1±1.3	84.0±1.1	81.5±2.0	84.0±1.2	89.7±0.40	84.5±2.3	89.7±0.40	92.7±1.1
Source-Risk	73.7±4.2	72.2±1.2	73.4±2.2	72.4±4.2	72.6±2.6	83.5±2.4	75.4±5.9	51.5±14	78.0±4.5	84.8±3.0	-	73.2±4.3	-
IWCV	73.3±4.3	72.3±4.6	74.1±0.93	74.0±3.0	76.6±4.5	76.4±6.3	74.5±4.3	54.3±12	76.9±3.7	86.2±1.5	-	76.4±6.3	-
DEV	73.3±4.9	73.1±9.2	72.1±4.4	72.1±3.4	71.7±7.0	69.1±12	68.9±17	54.6±15	69.5±9.3	75.6±4.4	-	72.6±12	-
DEV-N	71.8±6.0	70.8±3.5	71.7±2.2	70.9±2.7	74.1±6.1	82.9±2.3	74.8±2.5	66.5±9.7	73.6±1.6	85.1±3.5	-	75.8±3.3	-
Entropy	65.6±12	58.0±9.9	56.9±6.1	63.2±14	51.7±3.4	56.2±12	52.1±3.0	57.2±7.7	68.9±11	71.0±17	-	51.0±0.68	-
InfoMax	71.8±3.6	74.5±3.5	78.6±2.1	80.2±3.5	72.0±5.3	81.6±1.2	72.4±6.7	57.2±12	76.2±0.48	86.8±1.8	-	78.8±5.9	-
Corr-C	72.0±3.3	76.2±5.2	77.4±7.0	81.0±3.7	74.2±2.5	81.7±1.2	71.2±14	63.0±17	69.5±22	87.1±1.6	-	79.1±6.2	-
MCC (V)	70.0±7.1	71.9±1.1	74.5±1.2	78.0±4.9	69.5±5.1	73.8±14	64.5±16	54.9±11	74.6±2.6	85.0±3.3	-	73.8±14	-
BNM (V)	71.9±3.4	74.5±3.5	76.8±2.8	75.6±13	72.0±5.3	82.0±1.6	72.3±6.5	57.2±12	75.3±1.8	86.6±1.6	-	78.8±5.9	-
ClassAMI	55.5±2.9	65.7±10	78.3±2.6	77.8±1.1	58.9±11	82.8±3.2	74.4±8.5	68.5±6.2	75.5±5.7	82.1±7.8	-	74.2±8.7	-
SND	73.3±6.3	75.4±4.2	75.5±6.5	75.8±6.2	71.8±5.5	73.1±5.5	70.4±9.8	60.8±10	78.6±5.0	85.1±1.4	-	70.5±9.7	-
MixVal	72.7±12	75.4±9.4	74.7±11	79.7±2.8	71.7±12	70.2±12	68.8±9.8	51.5±9.8	73.5±1.4	87.9±1.7	-	48.5±6.0	-
TransScore	72.8±4.0	74.8±2.6	74.8±1.3	75.0±2.2	71.1±6.3	71.7±12	63.6±13	57.2±11	74.9±2.8	86.6±1.7	-	70.3±12	-

Table 16. Complete UDA results on RSNA→Child CXR (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 15. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	CoUDA	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	81.9±2.6	83.1±2.1	82.1±2.4	83.8±1.9	83.8±1.9	87.3±1.8	83.6±1.5	89.7±0.49	80.8±2.5	83.5±1.5	84.2±2.7	89.7±0.49	93.0±1.2
Source-Risk	73.4±5.6	72.3±1.7	74.4±2.8	71.0±5.4	72.4±3.7	84.4±2.9	75.0±8.0	84.9±4.0	52.3±15	77.5±6.0	-	72.4±6.0	-
IWCV	74.0±5.2	74.1±6.0	73.7±1.1	75.6±3.5	77.8±5.4	75.1±7.8	73.6±5.8	85.6±1.7	55.0±15	76.7±4.6	-	75.1±7.8	-
DEV	73.0±5.9	75.8±11	74.2±5.3	73.6±3.7	73.0±9.4	71.9±17	71.2±22	77.9±4.9	52.9±20	72.1±12	-	74.4±16	-
Entropy	68.8±12	51.1±9.9	56.7±6.9	63.6±17	50.0±3.9	51.2±14	51.1±3.6	79.7±18	55.6±7.7	72.1±13	-	51.1±0.92	-
InfoMax	71.4±4.9	74.7±4.3	78.5±2.8	80.8±4.9	75.2±5.8	81.7±1.5	70.5±8.6	85.9±2.0	62.6±15	76.1±0.51	-	81.2±7.2	-
Corr-C	71.8±4.5	79.5±5.2	79.5±8.3	80.6±4.7	75.2±3.0	81.8±1.5	72.8±17	86.3±1.6	64.6±23	80.0±26	-	81.7±7.3	-
MCC (V)	71.4±8.0	71.7±1.2	74.8±1.6	78.3±6.8	69.5±6.1	80.4±17	71.7±20	85.8±4.5	53.9±14	75.8±3.0	-	80.4±17	-
BNM (V)	71.4±4.6	74.7±4.3	77.6±3.5	80.8±17	75.2±5.8	82.0±2.1	70.5±8.3	85.9±1.8	62.6±15	75.8±2.4	-	81.2±7.2	-
SND	71.5±7.7	76.1±5.2	77.8±8.2	78.3±6.7	73.2±6.5	71.9±7.2	72.2±13	85.0±1.7	63.0±13	79.1±6.2	-	70.1±13	-
ClassAMI	56.8±3.4	62.6±13	77.8±3.5	77.8±1.5	53.6±14	84.2±3.9	76.9±9.6	83.9±10	65.3±7.6	75.8±7.6	-	78.2±9.7	-
DEV-N	73.4±6.4	72.2±4.2	72.9±2.5	70.4±3.5	72.0±7.2	83.1±3.1	74.9±3.1	87.0±4.0	68.4±13	73.0±1.8	-	75.4±4.0	-
MixVal	76.3±14	78.9±11	78.0±14	80.3±3.1	75.9±16	72.7±15	71.0±12	88.0±2.0	50.6±14	73.1±1.6	-	50.3±7.8	-
TransScore	73.3±5.6	74.7±3.5	74.8±1.4	74.5±2.7	68.8±8.3	76.0±15	68.5±15	87.7±1.9	62.6±14	75.8±3.4	-	72.3±15	-

Table 17. Complete UDA results on Child CXR→RSNA (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	75.1±0.36	74.6±0.99	77.2±1.1	76.2±0.32	78.1±0.45	77.3±0.66	77.2±0.47	76.3±0.89	77.0±0.39	73.2±0.44	76.4±1.5	78.3±0.26	82.4±0.46
Source-Risk	72.4±2.8	69.3±4.7	68.8±2.7	72.0±1.3	74.6±2.1	73.5±1.8	71.8±3.1	61.9±17	67.5±2.9	71.5±0.84	-	70.7±3.0	-
IWCV	73.5±1.5	71.7±3.8	70.5±3.0	70.2±1.2	76.1±1.4	70.1±7.5	69.3±4.5	59.2±12	70.9±3.5	71.9±0.64	-	58.7±11	-
DEV	73.5±1.5	66.6±5.8	69.3±3.7	68.2±4.6	75.7±1.0	74.3±2.4	67.7±9.6	53.5±18	67.9±6.6	71.9±0.64	-	63.3±18	-
DEV-N	71.7±2.8	68.4±4.4	68.8±2.7	71.5±1.2	74.9±2.1	73.5±1.8	71.8±3.1	59.4±19	67.5±2.9	71.5±0.84	-	70.6±3.6	-
Entropy	72.4±1.1	66.9±1.4	68.3±1.8	67.2±4.6	76.2±1.6	74.2±1.6	72.8±3.4	68.0±7.0	75.5±0.70	71.2±0.32	-	74.2±1.6	-
InfoMax	72.5±1.0	72.1±2.9	73.2±1.0	74.5±0.94	75.7±1.2	74.2±2.0	74.1±0.98	65.7±16	74.2±1.7	71.2±0.32	-	74.2±2.0	-
Corr-C	72.3±0.96	74.0±2.0	76.3±1.1	75.8±0.36	76.0±0.51	75.5±1.1	73.9±0.92	71.6±2.9	74.0±1.9	71.7±0.49	-	74.6±1.4	-
MCC (V)	72.6±1.2	67.5±2.1	68.8±1.8	69.1±3.4	74.3±4.3	74.8±2.1	74.4±1.4	72.5±2.8	75.7±0.99	71.2±0.32	-	72.0±1.3	-
BNM (V)	72.5±1.0	72.1±2.9	73.2±1.0	74.5±0.94	75.7±1.2	74.2±2.0	74.0±0.98	65.7±16	73.8±1.1	71.2±0.32	-	74.2±2.0	-
ClassAMI	71.4±1.2	69.2±1.8	69.5±1.7	70.9±2.5	75.3±0.84	73.8±1.4	72.9±1.5	70.4±3.2	70.0±3.7	71.3±0.36	-	72.9±2.3	-
SND	73.0±1.4	73.9±0.93	74.7±5.6	76.2±0.32	74.7±1.2	73.1±2.1	71.6±4.3	63.4±13	71.2±2.6	72.4±0.73	-	74.7±1.2	-
MixVal	72.6±1.1	68.0±2.3	70.3±2.6	71.7±3.3	76.7±0.72	73.7±0.94	73.2±1.1	61.6±16	73.3±1.1	71.2±0.28	-	73.3±1.1	-
TransScore	70.5±2.8	70.3±0.90	70.8±1.4	72.7±2.7	76.8±0.93	73.7±1.1	73.5±2.2	56.0±20	75.3±0.91	71.3±0.29	-	75.3±0.91	-

Table 18. Complete UDA results on Child CXR→RSNA (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 17. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	CoUDA	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	75.2±0.46	74.7±1.1	77.4±1.2	76.2±0.40	78.2±0.58	77.4±0.87	77.0±0.62	72.9±0.47	76.6±1.0	77.2±0.46	76.4±1.6	78.3±0.37	82.4±0.64
Source-Risk	73.5±3.4	72.2±5.1	70.1±3.0	72.5±1.3	73.7±2.5	73.5±2.5	73.3±3.8	71.9±1.0	71.4±20	67.4±3.5	-	71.3±3.9	-
IWCV	73.5±2.0	73.5±4.6	71.7±3.9	70.4±1.6	76.3±1.9	72.2±8.5	71.0±5.7	71.8±0.81	63.1±15	71.4±4.3	-	60.4±15	-
DEV	73.5±2.0	68.3±7.1	67.8±4.2	69.5±5.8	76.3±1.2	74.5±3.0	73.7±11	71.8±0.81	63.1±21	69.8±8.4	-	70.8±21	-
Entropy	72.5±1.3	66.2±1.7	69.2±2.0	65.6±5.5	76.8±1.8	74.1±2.1	73.2±4.2	71.2±0.42	69.2±8.5	75.5±0.82	-	74.1±2.1	-
InfoMax	72.6±1.3	71.1±3.2	73.2±1.3	74.8±1.2	74.9±1.3	74.0±2.5	74.2±1.3	71.2±0.42	73.2±18	74.1±2.3	-	74.0±2.5	-
Corr-C	71.7±1.1	74.7±2.4	76.7±1.3	75.6±0.36	75.9±0.61	76.0±1.3	74.1±1.1	72.0±0.57	71.8±4.0	73.4±2.4	-	74.9±1.9	-
MCC (V)	73.1±1.4	66.5±2.6	69.2±2.0	69.8±4.4	76.1±5.2	74.2±2.5	75.2±1.6	71.2±0.42	73.2±3.4	76.1±1.2	-	71.7±1.7	-
BNM (V)	72.6±1.3	71.1±3.2	73.2±1.3	74.8±1.2	74.9±1.3	74.0±2.5	74.1±1.3	71.2±0.42	73.2±18	74.1±1.5	-	74.0±2.5	-
SND	73.2±1.9	73.7±1.2	77.4±6.7	76.2±0.40	75.0±1.5	74.0±2.5	69.6±4.6	72.2±0.95	68.9±17	72.0±3.5	-	75.0±1.5	-
ClassAMI	71.3±1.4	69.2±2.4	70.3±2.1	71.5±3.1	75.2±1.1	73.5±1.8	72.1±1.8	71.2±0.43	71.8±3.5	70.6±4.4	-	73.0±2.4	-
DEV-N	70.5±3.4	67.9±5.1	70.1±3.0	70.6±1.3	75.9±2.2	73.5±2.5	73.3±3.8	71.9±1.0	71.4±21	67.4±3.5	-	72.4±4.0	-
MixVal	72.9±1.3	68.1±3.3	71.6±3.1	72.3±4.1	76.7±0.88	74.0±1.1	72.7±1.3	71.2±0.35	68.0±20	73.4±1.3	-	73.4±1.3	-

Table 19. Complete UDA results on LDD→CRD (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	79.0±0.82	82.1±0.52	82.4±0.54	82.6±0.42	83.0±0.52	82.4±0.90	83.2±0.50	83.0±0.50	83.0±0.63	81.7±0.80	82.6±0.47	83.6±0.13	95.7±0.34
Source-Risk	67.6±1.8	79.7±0.94	78.7±2.2	69.2±5.3	79.9±2.4	74.1±2.6	81.0±1.9	78.8±3.9	79.5±0.96	75.4±1.4	-	74.0±6.8	-
IWCV	69.5±4.2	79.5±0.43	77.7±3.4	71.8±3.6	79.2±2.1	74.7±2.2	79.7±3.1	75.4±6.5	73.8±5.4	76.1±1.1	-	74.7±2.2	-
DEV	73.4±6.4	77.9±3.3	79.8±0.45	77.6±2.1	79.8±3.8	72.9±8.8	76.2±4.7	54.6±12	79.8±3.2	79.3±1.6	-	74.8±10	-
DEV-N	68.2±2.0	79.9±0.72	78.2±1.3	69.8±5.2	80.1±2.5	74.1±2.6	81.4±1.1	78.8±3.9	79.2±1.3	75.5±1.3	-	73.9±6.8	-
Entropy	62.4±7.0	73.8±13	68.6±17	68.7±11	75.2±14	70.5±12	62.7±17	73.5±14	80.8±1.8	75.2±0.08	-	54.3±9.7	-
InfoMax	74.0±4.3	81.0±1.3	81.5±0.67	81.4±0.95	82.2±0.88	81.5±1.4	80.6±2.6	80.6±1.5	81.9±0.90	78.8±0.85	-	81.9±0.59	-
Corr-C	76.7±2.9	79.6±1.4	81.5±0.76	79.5±2.4	82.3±1.0	79.2±2.6	80.8±1.4	78.8±6.2	81.9±1.4	79.2±2.1	-	81.1±1.6	-
MCC (V)	67.4±0.93	79.8±0.76	79.9±1.6	74.4±1.3	81.3±1.6	75.7±5.2	81.7±0.49	79.9±3.2	81.1±1.9	75.2±0.08	-	75.7±5.2	-
BNM (V)	72.8±3.5	80.7±0.93	81.5±0.67	81.4±0.95	82.2±0.88	81.5±1.4	80.6±2.6	80.6±1.4	81.8±1.1	78.3±0.93	-	81.9±0.58	-
ClassAMI	69.5±7.8	76.8±2.7	80.8±1.2	72.5±6.2	79.3±2.6	78.5±3.9	79.2±1.9	80.7±1.2	76.6±4.2	77.2±1.6	-	78.6±3.2	-
SND	68.5±4.8	72.6±4.4	74.6±4.1	71.4±2.3	76.9±1.3	68.5±4.6	72.2±3.3	72.8±4.5	72.3±3.4	77.9±3.0	-	75.8±2.6	-
MixVal	69.3±8.8	80.7±0.95	80.9±0.85	74.6±5.2	80.7±1.6	75.9±6.5	78.5±1.6	71.2±22	79.0±4.2	76.6±0.63	-	79.0±4.2	-
TransScore	70.1±3.9	80.0±1.2	81.4±1.2	79.6±1.0	81.3±1.3	81.5±1.4	81.7±0.90	62.9±20	81.3±0.65	77.1±0.89	-	81.2±0.99	-

Table 20. Complete UDA results on LDD→CRD (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 19. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	CoUDA	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	79.1±1.1	82.2±0.62	82.3±0.66	82.5±0.58	83.3±0.55	82.5±1.2	83.4±0.64	82.1±0.87	82.8±0.56	82.8±0.73	82.7±0.46	83.6±0.15	96.0±0.36
Source-Risk	68.8±2.1	80.1±1.1	77.8±2.7	72.2±5.8	80.8±3.1	74.7±2.9	81.9±2.3	75.0±1.9	80.6±4.7	79.5±1.2	-	77.4±7.7	-
IWCV	70.6±4.7	79.7±0.56	77.8±4.5	72.5±4.8	80.0±2.6	76.0±2.7	81.1±3.8	75.4±1.2	77.7±8.2	74.7±7.3	-	76.0±2.7	-
DEV	72.8±7.9	79.3±4.0	79.8±0.61	77.9±2.6	81.3±4.5	75.9±11	77.6±6.3	79.8±2.0	48.3±12	80.8±3.8	-	80.0±12	-
Entropy	64.8±8.8	79.6±15	80.4±16	72.3±13	80.2±16	74.8±15	50.0±16	75.2±0.11	80.4±16	81.2±2.2	-	50.0±11	-
InfoMax	73.8±5.6	81.2±1.5	81.4±0.88	81.5±1.2	82.2±1.0	81.4±1.9	81.7±3.2	78.5±1.0	80.8±1.9	81.7±1.1	-	81.8±0.71	-
Corr-C	76.9±3.7	79.6±1.9	81.5±1.1	80.5±2.7	81.8±1.1	80.0±2.9	81.0±1.7	79.5±2.6	81.3±7.4	82.4±1.7	-	81.7±1.7	-
MCC (V)	67.8±1.1	79.6±0.98	80.4±2.0	74.7±1.7	80.7±1.7	77.8±6.4	81.6±0.63	75.2±0.11	81.0±3.9	82.0±2.2	-	77.8±6.4	-
BNM (V)	73.8±4.8	81.1±1.1	81.4±0.88	81.5±1.2	82.2±1.0	81.4±1.9	81.7±3.2	77.9±1.1	80.8±1.9	81.7±1.5	-	81.8±0.71	-
SND	68.5±6.1	74.7±4.6	76.4±5.0	70.9±3.1	76.6±1.8	69.5±5.7	71.5±4.0	78.1±3.9	72.8±5.5	70.8±4.2	-	76.3±3.6	-
ClassAMI	73.3±9.5	76.5±3.2	80.5±1.4	74.0±7.9	80.2±3.4	80.5±4.6	78.9±2.3	77.3±2.1	80.3±1.4	76.7±5.7	-	78.3±4.3	-
DEV-N	68.8±2.5	80.1±0.91	77.8±1.5	72.2±6.4	80.8±3.1	74.7±2.9	81.9±1.3	75.9±1.6	80.9±4.7	79.5±1.5	-	76.5±7.8	-
MixVal	68.2±11	80.7±1.3	81.4±0.91	74.3±6.5	80.9±2.0	77.8±7.5	79.1±2.1	77.0±0.73	81.0±25	80.6±5.1	-	80.6±5.1	-
TransScore	69.0±4.9	79.6±1.5	81.6±1.7	79.8±1.3	81.0±1.7	81.4±1.9	81.5±1.1	77.1±1.2	71.3±24	81.6±0.72	-	81.8±1.1	-

Table 21. Complete UDA results on CRD→LDD (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	77.2±3.0	78.6±1.6	77.9±5.8	81.0±1.5	77.7±1.9	78.5±3.6	80.0±3.6	74.8±3.4	76.3±5.0	63.9±1.3	76.5±4.8	82.7±1.1	98.7±0.17
Source-Risk	68.2±1.6	44.9±1.3	64.7±11	66.2±14	58.4±12	57.1±9.7	60.0±8.2	49.8±7.5	56.0±8.4	55.2±1.2	-	55.2±1.2	-
IWCV	67.7±7.3	46.1±3.6	61.7±15	70.0±10	56.5±6.4	59.2±11	58.3±11	62.0±17	57.8±8.0	54.5±1.9	-	63.7±6.1	-
DEV	65.9±13	45.3±5.9	65.7±12	63.0±7.3	62.0±11	63.9±11	55.7±8.1	48.3±15	58.5±8.2	52.4±2.3	-	59.9±9.7	-
DEV-N	68.2±1.6	51.5±13	67.7±13	63.5±15	56.4±12	57.1±9.7	59.3±9.2	74.1±4.3	56.0±8.4	55.2±1.2	-	55.2±1.2	-
Entropy	65.4±8.9	44.8±1.6	59.2±14	73.9±13	58.8±11	68.1±7.9	52.2±9.6	47.6±3.0	61.5±9.9	54.1±1.6	-	58.4±7.7	-
InfoMax	67.1±2.5	50.5±11	73.1±9.8	69.9±10	66.4±8.3	55.0±11	54.4±11	51.5±10	61.6±9.6	54.2±1.6	-	55.0±11	-
Corr-C	64.1±1.6	63.1±11	73.1±9.2	63.5±10	67.0±8.1	53.3±2.6	55.4±2.3	51.2±14	55.8±7.9	54.7±1.2	-	68.1±12	-
MCC (V)	68.0±2.1	44.8±1.6	59.8±15	70.0±12	62.6±11	63.0±14	55.1±12	48.6±4.6	49.4±3.7	54.2±1.6	-	63.0±14	-
BNM (V)	67.4±1.9	50.4±11	73.1±9.8	69.3±11	66.4±8.3	55.0±11	54.4±11	45.2±16	61.6±9.6	54.2±1.6	-	55.0±11	-
ClassAMI	68.2±3.9	50.7±11	66.2±16	66.7±8.0	62.4±11	52.7±9.3	58.9±6.6	47.1±14	64.2±1.8	50.7±3.6	-	73.1±8.4	-
SND	60.6±2.9	61.5±11	71.2±4.6	67.8±8.8	53.3±8.4	51.0±2.0	57.9±9.2	62.8±8.2	64.9±8.3	54.7±0.90	-	51.8±7.2	-
MixVal	67.7±1.8	47.7±9.4	71.7±11	64.6±11	61.8±13	63.0±13	54.2±15	37.7±6.5	67.2±2.4	56.0±2.2	-	67.2±2.4	-
TransScore	68.0±1.9	66.8±11	66.0±13	70.4±13	52.0±5.6	54.6±15	53.5±7.1	51.8±10	53.5±6.2	54.8±1.1	-	55.9±6.7	-

Table 22. Complete UDA results on CRD→LDD (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 21. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	CoUDA	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	76.7±4.1	77.7±1.9	80.1±7.0	81.4±1.9	78.2±2.3	76.5±3.8	80.0±4.6	64.1±1.7	75.5±4.0	76.1±6.6	76.6±5.1	82.7±1.4	98.8±0.20
Source-Risk	68.1±2.0	45.0±1.7	64.6±12	67.0±17	59.0±12	60.4±13	62.9±8.9	55.1±1.7	44.9±8.2	58.5±9.3	-	55.1±1.7	-
IWCV	65.4±9.5	45.7±4.5	52.9±15	70.9±13	59.5±7.9	62.4±14	64.7±12	54.4±2.5	70.8±18	58.0±11	-	64.7±8.3	-
DEV	71.9±15	43.3±6.1	71.5±14	61.4±8.4	58.6±12	62.4±14	54.0±8.6	52.8±3.1	51.8±20	62.9±9.7	-	58.6±12	-
Entropy	70.0±11	44.6±2.1	50.2±14	80.0±15	57.6±15	65.8±9.5	50.1±13	54.1±2.0	47.6±3.1	66.6±10	-	62.8±7.9	-
InfoMax	66.9±3.2	45.4±13	77.0±11	75.7±11	68.5±11	53.0±14	50.4±14	54.1±2.0	55.4±12	64.5±11	-	53.0±14	-
Corr-C	64.9±1.9	69.8±11	76.5±11	58.4±12	68.5±11	54.7±2.8	55.0±3.1	54.4±1.6	51.0±18	53.7±10	-	76.3±12	-
MCC (V)	68.1±2.3	44.6±2.1	50.2±15	73.8±13	62.4±14	62.8±19	51.9±13	54.1±2.0	47.6±5.5	47.6±4.4	-	62.8±19	-
BNM (V)	66.9±2.4	45.4±13	77.0±11	75.7±11	68.5±11	53.0±14	50.4±14	54.1±2.0	55.4±18	64.5±11	-	53.0±14	-
SND	60.0±3.4	59.8±14	72.2±5.6	69.2±11	52.6±10	51.0±2.8	53.0±11	55.0±1.0	65.3±10	64.7±11	-	52.6±10	-
ClassAMI	68.6±4.4	46.0±14	75.6±17	63.6±9.7	61.5±13	58.8±9.5	56.1±7.5	50.5±4.8	45.5±19	64.2±2.1	-	77.3±9.1	-
DEV-N	68.1±2.0	47.0±15	75.1±13	67.0±17	49.0±12	60.4±13	62.9±11	55.1±1.7	75.5±5.2	58.5±9.3	-	55.1±1.7	-
MixVal	67.4±2.1	43.4±11	75.9±13	61.8±13	68.3±13	65.4±18	50.1±20	56.3±2.9	36.8±8.0	66.3±2.6	-	66.3±2.6	-

Table 23. Complete UDA results on OCT→SLO (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	61.9±1.3	66.7±0.52	67.0±0.39	67.2±0.53	66.1±0.79	64.2±1.4	65.0±0.66	60.4±2.5	64.2±2.4	65.1±2.1	67.4±0.15	75.0±0.58
Source-Risk	59.5±1.1	63.3±1.6	66.3±0.59	64.0±0.65	63.8±1.2	57.6±3.9	55.8±5.0	38.1±4.1	54.2±3.5	-	57.6±13	-
IWCV	56.2±5.5	65.8±0.42	61.0±1.4	59.7±1.4	56.0±4.2	60.2±4.1	52.5±10	47.1±5.9	54.8±2.2	-	50.3±10	-
DEV	57.1±4.1	64.2±2.4	61.6±3.7	62.2±1.8	54.1±1.4	56.1±6.2	57.9±4.4	51.2±3.8	55.7±3.6	-	62.7±2.8	-
DEV-N	58.7±0.49	63.3±1.6	65.7±1.6	63.8±0.16	63.4±1.9	59.2±4.5	55.8±5.0	39.6±6.6	48.8±4.2	-	57.9±9.3	-
Entropy	58.0±3.3	63.2±1.5	66.1±0.53	63.6±1.6	64.0±0.60	50.4±3.9	50.1±0.71	45.4±4.6	48.8±4.8	-	50.2±3.9	-
InfoMax	59.2±1.2	63.7±1.9	65.8±0.23	66.4±0.58	64.0±0.60	54.8±7.9	56.4±6.4	44.6±3.9	49.5±4.2	-	49.7±4.0	-
Corr-C	58.9±1.9	63.8±2.0	65.8±1.1	65.9±0.74	64.5±0.21	59.7±3.0	54.5±5.1	41.5±4.7	53.5±4.3	-	57.3±7.0	-
MCC (V)	59.1±1.6	63.3±1.4	66.1±0.53	65.0±0.97	64.0±0.60	54.0±7.4	54.8±8.8	45.4±4.6	48.8±4.8	-	54.0±7.4	-
BNM (V)	59.2±1.2	63.4±1.6	66.2±0.43	66.4±0.58	64.0±0.60	52.2±5.3	56.4±6.4	44.6±3.9	49.5±4.2	-	49.7±4.0	-
ClassAMI	55.5±6.7	64.8±1.1	65.3±0.98	63.7±1.0	64.8±0.38	54.0±6.8	60.4±2.8	41.6±8.0	51.2±7.8	-	50.2±5.9	-
SND	58.9±1.8	62.7±1.0	62.5±0.94	61.6±2.9	59.4±0.87	60.8±3.8	57.0±10	55.3±8.5	60.3±8.0	-	59.4±0.87	-
MixVal	56.4±6.5	62.8±1.3	64.8±0.91	65.6±0.70	64.0±0.25	58.7±4.5	51.3±2.9	44.8±6.6	51.5±3.7	-	51.5±3.7	-
TransScore	59.2±1.2	64.4±0.54	65.7±1.1	65.2±0.94	64.2±0.20	56.8±4.2	47.3±2.3	46.3±4.0	54.5±3.8	-	54.5±3.8	-

Table 24. Complete UDA results on OCT→SLO (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 23. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	61.5±1.3	66.5±0.49	66.9±0.39	67.4±0.50	66.5±0.73	64.1±1.4	65.3±0.63	61.3±2.4	63.8±2.4	65.2±2.0	67.4±0.15	75.7±0.57
Source-Risk	59.1±1.0	62.9±1.6	66.5±0.54	63.6±0.56	63.7±1.2	55.5±3.4	58.4±4.5	37.2±4.0	55.4±3.4	-	63.6±12	-
IWCV	56.8±5.4	65.6±0.36	60.7±1.4	59.6±1.4	55.9±4.2	61.4±3.9	48.8±9.7	48.2±5.8	55.5±2.2	-	48.8±10	-
DEV	56.8±4.1	65.5±2.0	63.2±3.4	61.2±1.6	53.8±1.4	57.6±6.1	59.2±4.2	51.0±3.8	55.5±3.6	-	64.2±2.4	-
Entropy	59.7±3.0	63.9±1.4	65.9±0.49	64.1±1.5	64.0±0.60	50.6±3.9	50.0±0.70	42.9±4.1	46.7±4.4	-	50.0±3.9	-
InfoMax	59.4±1.2	64.3±1.8	65.9±0.23	66.1±0.52	64.0±0.60	56.2±7.8	58.6±6.1	43.5±3.8	48.8±4.2	-	48.8±3.9	-
Corr-C	59.8±1.8	64.8±1.8	66.2±1.0	66.0±0.73	64.4±0.20	61.3±2.6	54.8±5.1	41.5±4.7	54.4±4.2	-	61.3±6.1	-
MCC (V)	59.7±1.5	63.9±1.2	65.9±0.49	65.0±0.97	64.0±0.60	54.2±7.4	50.0±7.7	42.9±4.1	46.7±4.4	-	54.2±7.4	-
BNM (V)	59.4±1.2	64.2±1.5	66.1±0.41	66.1±0.52	64.0±0.60	54.1±5.0	58.6±6.1	43.5±3.8	48.8±4.2	-	48.8±3.9	-
SND	59.8±1.6	63.0±0.99	62.7±0.91	63.0±2.6	59.1±0.82	62.4±3.6	62.9±9.2	59.0±7.8	62.5±7.7	-	59.1±0.82	-
ClassAMI	58.7±6.1	65.0±1.1	65.1±0.97	63.4±0.97	64.7±0.37	56.9±6.3	58.8±2.5	37.2±7.1	47.9±7.2	-	47.9±5.6	-
DEV-N	58.7±0.49	62.9±1.6	66.5±1.4	63.8±0.15	63.7±1.9	57.9±4.4	58.4±4.5	37.2±6.3	50.2±4.0	-	62.9±8.2	-
MixVal	60.0±5.7	63.1±1.3	64.9±0.91	65.7±0.69	64.0±0.25	60.5±4.2	50.4±2.8	47.9±6.0	51.4±3.7	-	51.4±3.7	-
TransScore	59.6±1.2	64.3±0.53	66.0±1.1	65.5±0.90	64.3±0.20	55.9±4.1	46.0±2.0	44.5±3.6	55.5±3.7	-	55.5±3.7	-

Table 25. Complete UDA results on SLO→OCT (target accuracy, %; mean with std). Each cell reports the mean target accuracy (std) of the checkpoint a validator selects for an algorithm. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	60.7±1.2	63.0±1.0	64.6±1.5	65.4±0.34	63.6±1.1	62.3±1.4	61.3±3.3	58.4±5.2	62.0±1.6	62.6±2.0	65.6±0.62	74.4±0.38
Source-Risk	57.9±1.4	60.7±0.83	61.1±1.6	63.3±1.4	59.6±1.3	53.5±4.6	49.7±4.2	45.1±0.56	54.6±6.7	-	55.2±6.7	-
IWCV	56.6±0.78	56.2±0.90	60.5±2.5	60.8±0.25	58.7±0.39	54.1±3.6	48.6±1.2	51.2±3.5	55.7±4.7	-	52.0±5.4	-
DEV	58.4±1.7	56.8±3.4	61.4±2.3	60.3±1.8	56.1±1.9	55.9±3.4	51.8±4.9	46.2±1.5	57.5±3.0	-	54.2±7.8	-
DEV-N	57.0±3.0	61.6±2.3	62.4±0.69	62.2±1.6	60.3±1.7	52.6±6.2	49.4±4.5	43.7±2.5	49.7±3.8	-	60.1±4.9	-
Entropy	57.1±1.3	49.6±4.8	60.5±1.7	61.9±1.4	61.4±1.1	45.6±3.2	50.3±0.90	44.9±4.0	51.1±6.0	-	45.9±3.6	-
InfoMax	56.9±1.6	49.7±4.8	60.1±2.3	61.4±2.6	61.6±1.1	45.3±4.9	48.3±1.6	45.2±3.9	52.1±5.8	-	45.3±4.9	-
Corr-C	57.9±1.9	50.1±4.4	61.4±2.5	62.6±2.1	61.7±1.4	55.0±5.4	49.3±1.9	47.0±4.6	51.8±5.3	-	49.3±1.9	-
MCC (V)	56.9±1.5	49.6±4.8	60.5±1.7	61.0±1.2	61.4±1.1	45.5±3.1	49.7±1.6	44.9±4.0	50.8±5.6	-	45.5±3.1	-
BNM (V)	56.9±1.5	49.7±4.8	60.5±1.7	61.4±2.6	61.6±1.1	45.3±4.9	48.5±1.9	45.2±3.9	52.2±5.9	-	45.3±4.9	-
ClassAMI	57.8±2.1	59.8±1.5	59.5±2.5	60.4±1.1	61.8±1.6	52.6±7.5	47.0±1.4	43.6±3.8	52.7±6.0	-	46.0±5.9	-
SND	59.0±2.4	58.5±3.2	60.6±2.3	60.1±2.3	57.0±1.4	57.6±4.4	54.2±6.7	54.4±5.9	58.7±2.1	-	57.0±1.4	-
MixVal	58.1±1.9	49.8±4.4	59.7±2.1	62.3±2.2	60.0±1.4	46.2±4.6	50.7±2.7	44.8±3.5	53.6±5.6	-	52.2±6.5	-
TransScore	57.6±1.7	59.5±3.1	60.5±1.7	64.0±1.2	60.3±0.93	46.3±2.6	51.5±3.6	47.3±6.5	57.3±3.2	-	57.3±3.2	-

Table 26. Complete UDA results on SLO→OCT (target accuracy, %; median with 95% CI). Each cell reports the median target accuracy (95% confidence interval) of the checkpoint a validator selects for an algorithm, computed over the same runs as the mean±std results in Table 25. The per-algorithm columns report selection within a single algorithm, **Avg.** is their mean, and **Across-Algo** pools the checkpoints of all algorithms and selects across them. *Oracle* selects using target labels and is the actual best model, *SourceOnly* is trained on source data only, and *TargetOnly* is trained on labeled target data.

	SourceOnly	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	**Avg.**	**Across-Algo**	*TargetOnly*
Oracle	61.2±1.1	62.4±0.89	64.0±1.4	65.5±0.33	63.3±1.1	63.0±1.2	62.8±3.0	60.2±4.9	61.2±1.4	62.8±1.6	65.5±0.62	75.8±0.30
Source-Risk	58.1±1.4	61.1±0.76	61.6±1.5	62.7±1.3	59.3±1.3	53.1±4.6	48.8±4.1	45.2±0.54	53.2±6.6	-	54.2±6.7	-
IWCV	56.2±0.73	56.6±0.81	60.6±2.5	60.9±0.23	58.8±0.37	53.1±3.5	48.2±1.1	50.3±3.4	57.4±4.5	-	50.3±5.2	-
DEV	58.3±1.7	56.9±3.4	60.1±2.0	60.7±1.8	56.2±1.9	56.8±3.3	49.9±4.6	45.3±1.3	59.2±2.6	-	58.2±6.9	-
Entropy	56.4±1.2	48.4±4.7	60.6±1.7	62.4±1.4	60.9±1.1	44.2±3.0	50.0±0.85	46.6±3.7	49.8±5.9	-	44.2±3.3	-
InfoMax	56.2±1.5	48.5±4.7	60.6±2.3	60.4±2.4	61.5±1.1	44.5±4.8	47.5±1.4	47.1±3.6	54.5±5.4	-	44.5±4.8	-
Corr-C	58.6±1.8	48.5±4.2	60.5±2.4	62.4±2.1	60.9±1.3	53.5±5.2	48.5±1.8	44.5±4.0	53.2±5.2	-	48.5±1.8	-
MCC (V)	56.2±1.4	48.4±4.7	60.6±1.7	60.3±1.0	60.9±1.1	44.2±2.9	49.6±1.6	46.6±3.7	49.8±5.6	-	44.2±2.9	-
BNM (V)	56.2±1.4	48.5±4.7	60.6±1.7	60.4±2.4	61.5±1.1	44.5±4.8	47.5±1.7	47.1±3.6	54.5±5.5	-	44.5±4.8	-
SND	59.3±2.4	58.9±3.2	60.0±2.3	60.5±2.3	56.8±1.4	58.5±4.3	56.5±6.5	56.0±5.7	59.8±1.9	-	56.8±1.4	-
ClassAMI	58.7±2.0	60.4±1.4	58.3±2.2	59.8±0.94	61.0±1.4	52.2±7.5	46.9±1.3	42.5±3.6	50.4±5.6	-	45.2±5.9	-
DEV-N	58.1±2.8	61.1±2.2	62.4±0.68	62.9±1.5	61.1±1.5	53.7±6.2	47.7±4.3	44.5±2.5	50.1±3.7	-	62.4±4.4	-
MixVal	58.6±1.8	47.9±4.1	59.7±2.1	61.7±2.2	60.7±1.3	48.2±4.3	50.9±2.7	45.3±3.5	52.4±5.5	-	48.8±5.8	-
TransScore	58.0±1.7	59.5±3.1	60.6±1.7	64.5±1.1	60.7±0.87	47.3±2.5	51.9±3.6	44.8±6.2	56.5±3.2	-	56.5±3.2	-

Table 27. Within-algorithm Spearman's ρ on ADNI-1→ADNI-2. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A
Source-Risk	0.80	0.76	0.74	0.74	0.74	0.70	0.53	0.79	0.80
IWCV	0.49	0.29	0.31	-0.28	0.29	0.04	0.05	0.48	0.38
DEV	0.56	0.49	0.47	0.04	0.24	0.21	0.19	0.57	0.57
DEV-N	0.77	0.74	0.72	0.71	0.73	0.69	0.53	0.78	0.79
Entropy	0.45	0.61	0.60	0.57	0.55	0.51	0.33	0.72	0.67
InfoMax	0.78	0.87	0.87	0.87	0.83	0.79	0.61	0.81	0.89
Corr-C	0.82	0.79	0.81	0.83	0.78	0.74	0.62	0.82	0.87
MCC (V)	0.70	0.83	0.82	0.81	0.76	0.73	0.55	0.78	0.86
BNM (V)	0.75	0.86	0.86	0.85	0.82	0.79	0.60	0.80	0.88
SND	0.75	0.58	0.57	0.65	0.50	0.45	0.38	0.78	0.66
ClassAMI	0.29	0.70	0.70	0.73	0.69	0.67	0.53	0.47	0.71
MixVal	0.22	0.33	0.37	0.33	0.35	0.35	0.25	0.31	0.29
TransScore	0.74	0.86	0.85	0.84	0.80	0.77	0.59	0.78	0.88

Table 28. Within-algorithm Spearman's ρ on ADNI-1→ADNI-3. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A
Source-Risk	0.52	0.52	0.61	0.40	0.47	0.40	0.48	0.72	0.75
IWCV	0.16	0.14	0.22	-0.12	0.15	0.11	0.13	0.50	0.36
DEV	0.22	0.26	0.35	0.11	0.18	0.22	0.03	0.49	0.48
DEV-N	0.51	0.52	0.61	0.39	0.48	0.40	0.28	0.70	0.75
Entropy	0.35	0.53	0.44	0.22	0.20	0.24	0.05	0.55	0.56
InfoMax	0.46	0.61	0.58	0.39	0.39	0.18	0.43	0.66	0.73
Corr-C	0.40	0.35	0.37	0.21	0.29	0.03	0.20	0.64	0.64
MCC (V)	0.45	0.66	0.64	0.41	0.40	0.30	0.42	0.64	0.70
BNM (V)	0.46	0.63	0.61	0.40	0.39	0.20	0.42	0.64	0.72
SND	0.34	0.14	0.24	0.18	0.24	-0.01	0.24	0.62	0.52
ClassAMI	0.02	0.56	0.58	0.37	0.46	0.41	0.44	0.41	0.61
MixVal	0.13	0.20	0.27	0.13	0.20	0.19	0.26	0.27	0.29
TransScore	0.42	0.60	0.59	0.37	0.38	0.16	0.41	0.55	0.71

Table 29. Within-algorithm Spearman's ρ on ADNI-2→ADNI-1. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A
Source-Risk	0.73	0.58	0.53	0.59	0.58	0.56	0.68	0.76	0.79
IWCV	0.37	-0.04	0.10	-0.28	0.39	0.16	0.03	0.45	0.30
DEV	0.40	0.32	0.33	0.17	0.30	0.22	0.11	0.47	0.47
DEV-N	0.70	0.61	0.58	0.62	0.61	0.60	0.68	0.76	0.81
Entropy	0.32	0.51	0.44	0.45	0.34	0.26	0.38	0.67	0.61
InfoMax	0.77	0.76	0.69	0.78	0.70	0.69	0.79	0.79	0.88
Corr-C	0.79	0.73	0.60	0.71	0.66	0.67	0.72	0.82	0.85
MCC (V)	0.69	0.71	0.66	0.74	0.56	0.57	0.76	0.77	0.86
BNM (V)	0.74	0.75	0.69	0.77	0.70	0.68	0.79	0.78	0.87
SND	0.63	0.31	0.18	0.37	0.42	0.49	0.38	0.71	0.61
ClassAMI	0.31	0.68	0.59	0.67	0.62	0.63	0.62	0.58	0.75
MixVal	0.24	0.25	0.25	0.36	0.29	0.26	0.55	0.30	0.35
TransScore	0.75	0.75	0.68	0.77	0.69	0.65	0.75	0.77	0.87

Table 30. Within-algorithm Spearman's ρ on ADNI-2→ADNI-3. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A
Source-Risk	0.72	0.68	0.66	0.69	0.67	0.57	0.57	0.74	0.82
IWCV	0.38	0.30	0.23	-0.23	0.20	0.01	0.24	0.54	0.60
DEV	0.43	0.48	0.36	0.18	0.36	0.29	0.36	0.68	0.68
DEV-N	0.64	0.61	0.57	0.58	0.58	0.49	0.52	0.71	0.78
Entropy	0.30	0.20	0.15	0.17	0.19	0.19	0.19	0.63	0.49
InfoMax	0.76	0.71	0.68	0.68	0.64	0.51	0.36	0.63	0.80
Corr-C	0.63	0.51	0.55	0.56	0.48	0.33	0.19	0.61	0.58
MCC (V)	0.70	0.64	0.60	0.62	0.60	0.57	0.41	0.68	0.73
BNM (V)	0.75	0.72	0.68	0.69	0.65	0.54	-0.04	0.55	0.78
SND	0.53	0.39	0.45	0.49	0.35	0.24	0.17	0.52	0.44
ClassAMI	-0.01	0.66	0.57	0.56	0.65	0.60	0.55	0.23	0.55
MixVal	0.37	0.29	0.27	0.32	0.30	0.23	0.24	0.28	0.31
TransScore	0.75	0.72	0.69	0.67	0.64	0.53	0.39	0.64	0.78

Table 31. Within-algorithm Spearman's ρ on ADNI-1+2→AIBL. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	AD2A
Source-Risk	0.71	0.56	0.65	0.56	0.62	0.54	0.58	0.66	0.77
IWCV	0.52	0.15	0.18	-0.28	0.25	0.08	0.22	0.41	0.53
DEV	0.60	0.45	0.37	0.21	0.43	0.30	0.43	0.51	0.67
DEV-N	0.71	0.56	0.62	0.56	0.62	0.56	0.57	0.67	0.76
Entropy	0.55	0.46	0.46	0.41	0.50	0.41	0.43	0.72	0.69
InfoMax	0.65	0.40	0.58	0.46	0.45	0.31	0.54	0.75	0.71
Corr-C	0.42	-0.04	0.24	0.06	0.04	-0.12	-0.09	0.41	0.35
MCC (V)	0.65	0.54	0.62	0.54	0.58	0.44	0.48	0.72	0.76
BNM (V)	0.66	0.47	0.61	0.50	0.50	0.35	-0.13	0.51	0.73
SND	0.34	-0.14	0.09	0.01	-0.11	-0.20	-0.18	0.24	0.16
ClassAMI	-0.25	0.37	0.38	0.33	0.46	0.50	0.39	0.29	0.60
MixVal	0.31	0.27	0.21	0.13	0.16	0.13	0.18	0.24	0.27
TransScore	0.60	0.42	0.57	0.44	0.42	0.29	0.34	0.64	0.69

Table 32. Within-algorithm Spearman's ρ on RSNA→Child CXR. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA
Source-Risk	0.11	0.20	-0.08	0.21	0.49	0.22	-0.10	0.47	0.57
IWCV	-0.05	0.14	-0.06	-0.02	0.41	0.17	0.03	0.30	0.42
DEV	-0.05	0.06	-0.13	-0.08	0.26	0.10	0.11	-0.01	0.24
DEV-N	-0.20	-0.17	-0.30	0.23	0.39	0.22	-0.08	0.17	0.46
Entropy	-0.16	-0.15	-0.39	-0.19	0.33	0.07	-0.14	0.26	0.34
InfoMax	0.29	0.32	0.01	0.09	0.65	0.58	0.01	0.52	0.56
Corr-C	0.75	0.75	0.63	0.18	0.58	0.59	0.07	0.70	0.65
MCC (V)	0.10	0.11	-0.22	-0.08	0.45	0.23	-0.14	0.33	0.47
BNM (V)	0.24	0.27	-0.04	0.08	0.65	0.57	0.01	0.49	0.55
SND	0.76	0.76	0.68	0.24	0.39	0.50	0.16	0.62	0.59
ClassAMI	-0.18	0.64	0.44	-0.28	0.57	0.44	0.04	0.07	-0.25
MixVal	0.43	0.37	0.46	0.36	0.30	0.12	-0.10	0.37	0.15
TransScore	0.05	0.09	-0.15	0.13	0.63	0.58	-0.02	0.24	0.54

Table 33. Within-algorithm Spearman's ρ on Child CXR→RSNA. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA
Source-Risk	0.19	0.11	-0.08	0.22	0.12	0.17	-0.08	0.02	-0.11
IWCV	0.20	0.16	0.12	0.14	0.23	0.09	-0.00	0.14	0.24
DEV	0.18	0.17	0.11	0.15	0.24	0.10	-0.02	0.11	0.23
DEV-N	0.19	0.10	-0.07	0.22	0.12	0.18	-0.08	0.03	-0.10
Entropy	-0.12	-0.29	-0.25	0.44	0.07	0.32	-0.01	0.60	-0.43
InfoMax	0.81	0.79	0.87	0.47	0.58	0.44	0.09	0.73	-0.36
Corr-C	0.96	0.97	0.99	0.43	0.61	0.38	0.13	0.69	-0.16
MCC (V)	-0.01	-0.15	0.04	0.42	0.15	0.34	-0.01	0.61	-0.42
BNM (V)	0.76	0.72	0.84	0.47	0.57	0.44	0.09	0.73	-0.37
SND	0.90	0.92	0.96	-0.27	0.58	0.29	0.15	0.46	0.58
ClassAMI	0.35	-0.08	0.16	0.21	0.05	0.12	0.12	0.45	0.06
MixVal	0.07	0.24	0.44	0.39	-0.19	-0.16	0.11	0.18	-0.14
TransScore	0.22	-0.05	-0.08	0.18	0.40	0.39	0.06	0.67	-0.37

Table 34. Within-algorithm Spearman's ρ on LDD→CRD. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA
Source-Risk	0.57	0.58	0.06	0.68	0.20	0.74	0.30	0.63	-0.08
IWCV	0.43	0.22	-0.19	-0.03	-0.18	-0.25	-0.02	-0.16	-0.16
DEV	0.43	0.28	-0.10	0.14	-0.09	-0.11	0.02	0.13	-0.09
DEV-N	0.55	0.55	0.04	0.66	0.18	0.72	0.28	0.57	-0.10
Entropy	0.53	0.57	-0.06	0.68	0.07	0.67	0.26	0.69	-0.07
InfoMax	0.78	0.85	0.85	0.87	0.89	0.84	0.39	0.93	0.40
Corr-C	0.65	0.75	0.87	0.81	0.87	0.76	0.39	0.83	0.84
MCC (V)	0.63	0.66	0.08	0.78	0.20	0.77	0.34	0.76	0.05
BNM (V)	0.77	0.84	0.81	0.86	0.87	0.84	0.39	0.93	0.35
SND	0.16	0.34	0.70	0.09	0.74	0.34	0.11	0.44	0.67
ClassAMI	0.30	0.15	0.23	0.36	0.26	0.68	0.45	0.48	0.38
MixVal	0.66	0.58	0.43	0.68	0.23	-0.08	0.24	0.42	0.42
TransScore	0.48	0.58	0.20	0.68	0.73	0.80	0.26	0.74	0.03

Table 35. Within-algorithm Spearman's ρ on CRD→LDD. Each cell is the mean ρ over five folds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD	CoUDA
Source-Risk	-0.01	0.44	0.42	0.11	0.46	0.15	-0.04	-0.18	0.25
IWCV	0.28	0.13	0.19	0.09	0.10	0.13	0.15	0.17	0.20
DEV	0.22	0.17	0.24	0.17	0.16	0.12	-0.15	0.15	0.20
DEV-N	0.00	0.45	0.43	0.12	0.47	0.16	0.96	-0.18	0.26
Entropy	-0.21	0.34	0.28	-0.02	0.46	0.06	-0.07	-0.23	0.20
InfoMax	-0.02	0.49	0.47	0.15	0.03	0.31	0.01	-0.34	0.20
Corr-C	0.39	0.58	0.43	0.38	-0.13	0.35	0.05	-0.42	0.18
MCC (V)	-0.14	0.39	0.34	0.05	0.47	0.10	-0.05	-0.24	0.19
BNM (V)	-0.05	0.48	0.45	0.14	0.06	0.29	-0.02	-0.33	0.20
SND	0.50	0.21	0.15	0.37	-0.25	0.32	0.12	-0.14	-0.21
ClassAMI	-0.10	0.44	0.34	0.19	0.32	0.17	-0.18	-0.15	-0.08
MixVal	-0.12	0.32	0.29	0.06	0.38	-0.02	-0.18	0.01	0.18
TransScore	-0.14	0.30	0.28	0.04	0.12	0.24	0.02	-0.24	0.16

Table 36. Within-algorithm Spearman's ρ on OCT→SLO. Each cell is the mean ρ over three random seeds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD
Source-Risk	-0.08	0.40	0.27	0.59	-0.15	-0.12	-0.29	0.15
IWCV	0.06	-0.10	-0.06	-0.61	0.22	-0.02	0.02	-0.04
DEV	0.02	-0.06	0.02	-0.41	0.16	-0.09	0.07	-0.13
DEV-N	-0.05	0.42	0.21	0.58	-0.16	-0.12	-0.27	0.07
Entropy	-0.15	0.44	0.22	0.76	-0.11	-0.16	-0.34	0.15
InfoMax	-0.11	0.48	0.60	0.78	0.45	-0.04	-0.37	0.32
Corr-C	0.15	0.46	0.70	0.78	0.52	0.05	-0.26	0.37
MCC (V)	-0.14	0.45	0.26	0.77	-0.02	-0.14	-0.35	0.18
BNM (V)	-0.11	0.48	0.56	0.78	0.44	-0.05	-0.37	0.32
SND	0.26	-0.12	0.14	0.13	0.51	0.22	-0.02	0.34
ClassAMI	0.12	0.44	0.29	0.26	0.18	0.07	-0.06	-0.10
MixVal	-0.13	0.48	0.26	0.72	0.27	-0.07	-0.28	0.22
TransScore	-0.16	0.32	0.16	0.64	0.19	-0.16	-0.35	0.32

Table 37. Within-algorithm Spearman's ρ on SLO→OCT. Each cell is the mean ρ over three random seeds between the validation score and target accuracy across checkpoints within a single algorithm. Positive values indicate ranking in the intended direction.

	MMD	DANN	CDAN	DALN	MCC	BNM	ATDOC	MCD
Source-Risk	0.16	0.36	0.49	0.53	-0.35	-0.09	-0.26	0.19
IWCV	0.01	-0.06	-0.03	-0.40	0.15	0.02	0.08	0.04
DEV	-0.11	0.19	0.10	-0.00	0.01	0.07	0.05	-0.04
DEV-N	0.12	0.33	0.37	0.54	-0.34	-0.07	-0.26	0.18
Entropy	0.04	0.35	0.51	0.65	-0.39	-0.18	-0.44	0.13
InfoMax	0.09	0.36	0.59	0.68	-0.18	-0.07	-0.43	0.18
Corr-C	0.07	0.31	0.41	0.69	-0.06	-0.05	-0.33	0.26
MCC (V)	0.08	0.36	0.53	0.67	-0.40	-0.15	-0.44	0.14
BNM (V)	0.08	0.36	0.59	0.68	-0.19	-0.07	-0.43	0.18
SND	0.07	-0.02	-0.07	0.12	0.08	0.06	0.01	0.18
ClassAMI	-0.36	0.41	0.47	0.60	-0.23	-0.18	-0.26	-0.03
MixVal	0.08	0.35	0.50	0.61	-0.19	-0.13	-0.19	0.34
TransScore	0.05	0.31	0.48	0.17	-0.32	-0.07	-0.34	0.24